



Pre installation

Installation

Operating

**LC 10MT Chilled Water System for
GE Signa 1.5T TwinSpeed MRI System**

**Technical Publication 473
Revision 1**

The People of Ellis and Watts are committed to set goals and objectives to meet the expectations and needs of our customers. Responsibility and ownership at the point of operation will enhance the success of our mission

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REVISION HISTORY PAGE

REVISION NUMBER	DATE	REASON FOR CHANGE	AUTHOR
A	07/20/01	INITIAL DRAFT	S. FUTRAL
PRELIMINARY	08/09/01	PRELIMINARY	S. FUTRAL
0	9/20/01	INITIAL RELEASE	S. FUTRAL
1	11/28/01	REVISED SHIP LOOSE BOX SIZE IN TABLE 2, CHANGED FI DRAWING NUMBER ON PAGE 24 FROM FI-0015 TO FI-0097	S. FUTRAL

1 INTRODUCTION

1-1 CONTACT ELLIS & WATTS

If a non-service issue arises concerning the equipment or the documentation provided with the chiller you can contact Ellis & Watts in writing at the following address, by phone, or via E-mail at the Ellis & Watts web page.

**Ellis & Watts International, Inc.
Thermal Sales
4400 Glen Willow Lake Lane
Batavia, OH 45103**

Phone: 513-752-9000

Internet: WWW.ELLISWATTS.COM

Ellis & Watts provides 24-hour service support in the event that the Ellis and Watts equipment fails to perform properly. Service calls are processed more easily if you provide the service support department with the chiller **model number, serial number, the site name and location, and a description of the problem including the alarm or fault displayed on the operator interface**. The service support department will dispatch a qualified local service technician to correct the problem.

The 24 hour service support department phone numbers are:

**Domestic Toll Free: 1-888-744-3195
International: 513-768-3195**

1-2 WARRANTY REPLACEMENT PART PROCEDURE

To initiate a warranty claim with Ellis & Watts, contact the Thermal Sales Department. The equipment model number and serial number are required to process a warranty claim. The Thermal Sales representative will issue a Return Goods Authorization (RGA) number. The RGA number must be on the returned part's package.

NOTICE

Items returned without an RGA number will not be processed until the necessary information is obtained from the sender.

1-3 INSTALLATION TOOLS & EQUIPMENT

Item Description	QTY	Used For	Supplied By
Prybar, 2 foot (.61m)	1	Uncrating Equipment	Installer
Prybar, lever, 5000 lb. (2250kg)	1	Lifting Equipment	Installer
Razor Knife	1	Cutting Hose	Installer
10 foot (3.05 m) Stepladder	2	Uncrating Equipment	Installer
Forklift, 2 ton (1814 kg) Minimum	1	Lifting and Moving TGWC and Associated Equipment	Installer
Metal Snips	1	Cut banding Straps	Installer
2 Eye, Lift Straps 2 inch W x 18ft. Lg. (5.08cm W x 5.49m Lg.)	2	Lifting TGWC off Shipping Skid	Installer
C Clamp, 10 inch (25.4cm)	2	Securing Straps	Installer
Drill 3/8 inch minimum	1	Drill mounting holes	Installer
4 Conductor power cable Wire size and length determined by installer	1	Power for TGWC	Installer
3/8 inch (M10) Hardware Length to be determined	4	For tie down bracket	Installer
Light oil or soapy water	1 oz	Lubricating Hose Barbs	Installer
Loctite 567 or equivalent	4 oz	Sealing Pipe Threads	Installer

Table 1 **Installation Tools And Supplies**

1-3-1 Installation Responsibility Matrix

	Cust. Contr.	GE Service	E&W Service	Who Pays
Once Cooling Cabinets Arrive				
Unload TGWC from Truck. See Section 4-0	X			Customer
Move TGWC to equipment room and mount (Use Seismic Tie Down Brackets where required by law). See Section 5-1	X			Customer
Connect customer supplied power cable from MDP to TGWC. See Section 5-3	X			Customer
Install anti tipping legs. See Section 5-2	X			Customer
Once MR System Arrives				
Install water lines from TGWC to Gradient Coil. See Section 5-3.		X		GE
Fill TGWC with Propylene Glycol. See Section 6-1.		X		GE
Start up TGWC and ensure proper phasing and inspect for leaks. See Section 6-1.		X		GE
Verify operation per E&W checklist when load is available. See Section 6-1.			X	E&W

1-4 EXPLANATION OF NOTES, CAUTIONS, AND WARNINGS

A **NOTE** is used to provide supplemental information or direction.

CAUTION

A **CAUTION** is used to call particular attention to a step of a procedure, which, if not strictly followed, could result in damage to, or destruction of equipment.

WARNING

A **WARNING** is used to call attention to a particular step of a procedure, which, if not strictly followed, could result in serious injury, or death of personnel.

1-5 SAFETY WARNINGS**WARNING**

NO PERSON SHALL REACH INTO THE UNIT ENCLOSURE, WHICH CONTAINS LIVE ELECTRICAL CIRCUITS AS WELL AS ROTATING PARTS, FOR THE PURPOSE OF INSPECTING, SERVICING, ADJUSTING, OR MAINTENANCE EXCEPT IN THE PRESENCE OF SOMEONE CAPABLE OF RENDERING AID. CONTACTING SUCH ITEMS COULD CAUSE SERIOUS BODILY INJURY AND OR DEATH.

WARNING

PERSONNEL WORKING NEAR HIGH VOLTAGES SHOULD BE IN THE PRESENCE OF SOMEONE HAVING RESUSCITATION SKILLS. COMING IN CONTACT WITH HIGH VOLTAGE COULD CAUSE SERIOUS BODILY INJURY AND OR DEATH.

WARNING

TURN OFF AND TAG OUT THE MAIN POWER AT THE DISCONNECT, DURING ANY SERVICING OR MAINTENANCE OPERATIONS. REMOTE STARTING COMPONENTS OR DIRECT CONTACT WITH ELECTRICAL COMPONENTS MAY RESULT IN INJURY TO PERSONNEL.

WARNING

TURN OFF AND TAG OUT THE DISCONNECT TO THE UNIT BEFORE WORKING ON CONTACTORS OR OTHER COMPONENTS IN THE ELECTRIC BOX. WHEN THE MAIN POWER SUPPLY TO THE UNIT IS ON, THE LIVE SIDE OF EACH CONTACTOR CONTAINS HIGH VOLTAGE REGARDLESS OF ANY OTHER SWITCH SETTING. DO NOT ATTEMPT MAINTENANCE PROCEDURES INSIDE EQUIPMENT WITH HIGH VOLTAGE ON. REMOVE POWER AND TAG OUT THE DISCONNECT. GROUND ELECTRICAL COMPONENTS PRIOR TO TOUCHING TO PREVENT BODILY INJURY.

WARNING

NO SMOKING OR OPEN FLAMES WITHIN 50 FEET (15 M) OF REFRIGERANT SYSTEM. REFRIGERANT COMING IN CONTACT WITH LIT CIGARETTES CAN CREATE AN IRRITATING AND POSSIBLY TOXIC GAS.

SAFETY WARNINGS CONTINUED**WARNING**

STAND CLEAR DURING LIFTING OPERATIONS. OPERATOR ERROR AND/OR EQUIPMENT MALFUNCTION CAN CREATE OPPORTUNITIES FOR PERSONAL INJURY.

WARNING

THE UNIT HAS A CLOSED LOOP HIGH PRESSURE REFRIGERANT SYSTEM. DO NOT CLOSE, OPEN OR ADJUST VALVES IN THE REFRIGERANT SYSTEM UNLESS A SPECIFIC INSTRUCTION STATES THAT VALVES MUST BE OPENED, CLOSED OR ADJUSTED. PERSONNEL COULD BE INJURED OR EQUIPMENT COULD BE DAMAGED.

WARNING

BEFORE BRAZING OR SOLDERING ANY REFRIGERATION COMPONENT, RECOVER REFRIGERANT PER US ENVIRONMENTAL PROTECTION AGENCY (EPA) APPROVED PROCEDURE TO RELEASE THE PRESSURE FROM THE REFRIGERATION SYSTEM. SUDDEN DECOMPRESSION OF REFRIGERANT PIPING CAN CAUSE SERIOUS PERSONAL INJURY.

WARNING

USE EXTREME CARE WHEN BRAZING OR SOLDERING IN THE PRESENCE OF R-134a. WHEN R-134a COMES IN CONTACT WITH HOT METAL, AN IRRITATING GAS CAN FORM. HAVE SUFFICIENT FRESH AIR AND VENTILATION WHEN BRAZING, SOLDERING OR USING A HALIDE TORCH LEAK DETECTOR.

WARNING

THE COMPRESSOR MUST ALWAYS BE OPERATED WITH THE COMPRESSOR DISCHARGE SERVICE VALVE IN THE BACK-SEAT (FULLY OPEN) POSITION. FAILURE TO DO SO CAN CAUSE SERIOUS COMPONENT FAILURE.

WARNING

WHEN AN ABNORMAL CONDITION IS NOTED AND PROCEDURES IN THIS MANUAL DO NOT SPECIFICALLY RELATE TO THE NOTED CONDITION, WORK SHALL BE STOPPED. TECHNICALLY QUALIFIED PERSONNEL SHALL GIVE GUIDANCE ON THE ABNORMAL CONDITION BEFORE WORK CAN BE CONTINUED, TO PREVENT POSSIBLE PERSONAL INJURY OR EQUIPMENT DAMAGE.

2 SHIPPING AND DELIVERY DATA

The General Electric Medical Systems (GEMS) part number 2298100-3 High Voltage Indoor Air Cooled Unit (380V – 50 Hz , 480V – 60 Hz) is the Ellis and Watts chiller model LC10MT. The LC10MT consists of a liquid chiller referred to as Twin Gradient Water Cooler (TGWC).

2-1 SHIPMENT

International shipment shall be shipped to a collection point where all components for the magnet will be collected and shipped to the construction site.

Unit	Height	Width	Depth	Approximate Weights
TGWC Indoor Domestic	76 inches (185.88 cm)	34 inches (59.38 cm)	40 inches (76.20 cm)	830 lbs. (377 kg)
TGWC Indoor International	82 inches (208.28 cm)	40 inches (101.6 cm)	46 inches (116.84)	918 lbs. (417.27 kg)
Ship Loose Box	14.5 inches (36.83 cm)	26 inches (66.04 cm)	56 inches (142.24 cm)	150 lbs. (330 kg.)

Table 2 **Shipping Dimensions**

2-2 STORAGE REQUIREMENTS

The non-operation environmental conditions are as follows:

ITEM	REQUIREMENT
Ambient Temperature	-40°F (-40°C) to 158°F (70°C)
Humidity	5% to 100% RH
Altitude	984 feet (300 m) below sea level 17,060 feet (5,200 m) above sea level
Acceleration	+/- 1.5 G any axis
Magnetic Field	30 gauss
NOTE: Do not expose to water (rain, snow, salt air)	

Table 3 **Environmental Conditions**

NOTE: The TGWC should be kept in its shipping crate until it has been moved to the installation site.

3 PRE-INSTALLATION REQUIREMENTS

3-1 EQUIPMENT PLACEMENT

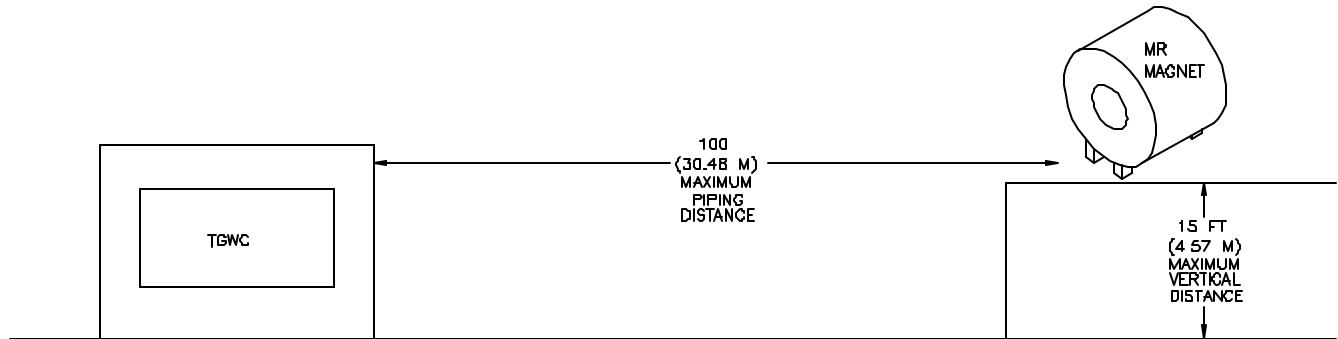


Illustration 1 **PLACEMENT OF TGWC TO MAGNET**

The maximum piping distance from the TGWC to the magnet is 100 feet (30.48 m), which includes a vertical maximum of 15 feet (4.57m).

3-2 SITE ENVIRONMENT

3-2-1 Operating Specifications

ITEM	REQUIREMENT
Ambient Temperature	41°F to 104°F (5°C to 40°C)
Humidity	10% to 90% RH
Altitude	984 feet (300m) below sea level to 11,800 feet (3,597m) above sea level
Magnetic Field	30 gauss
Heat Output to Environment	18.0 kW
Noise Levels	75 dBa measured one meter in any direction

Table 4 **Operating Specifications**

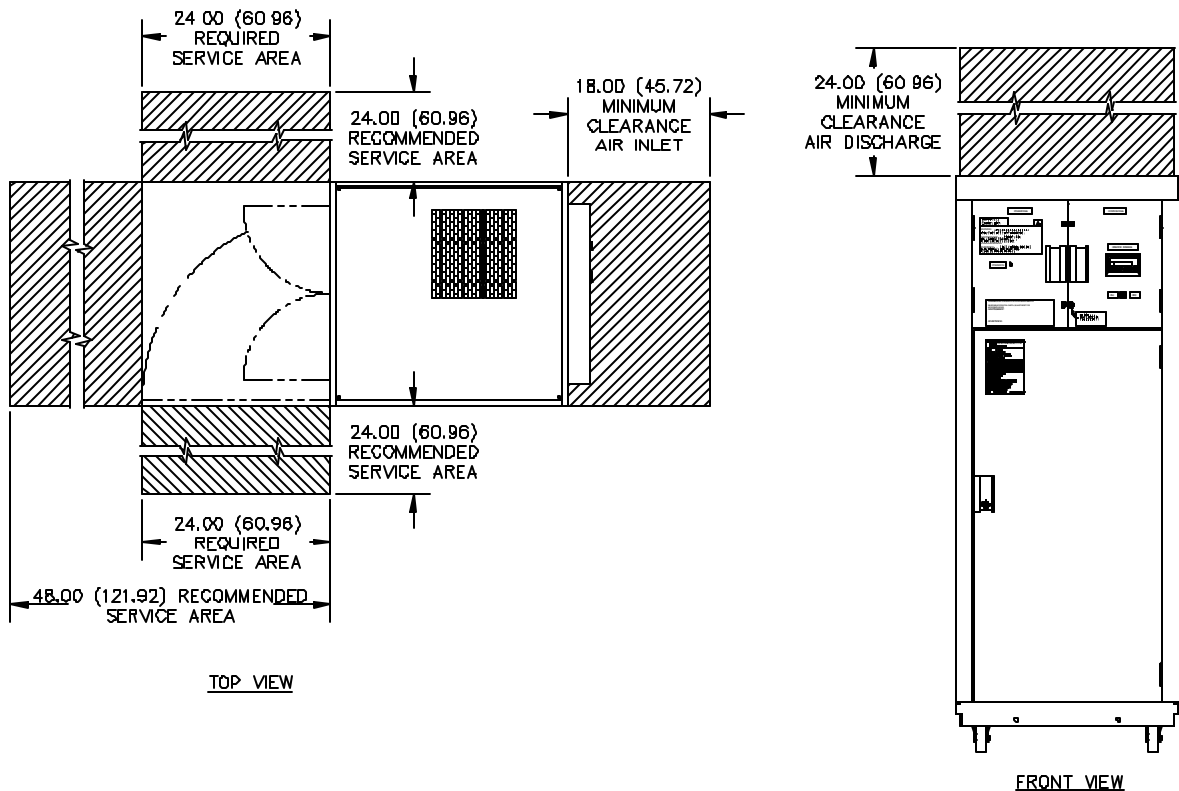
3-2-2 Power Requirements

ITEM	REQUIREMENT
Wire – <u>NOTE</u>	4AWG (25mm ²) maximum 8AWG (10 mm ²) minimum 4 Conductor with ground
Power Consumption	11.55 KVA (maximum)
Voltage Requirements	400 VAC $\pm$ 10% @ 50 Hz $\pm$ 3 Hz 480 VAC $\pm$ 10% @ 60 Hz $\pm$ 3 Hz
Circuit Size	50 AMP (maximum)
<u>NOTE:</u> The TGWC will be hardwired to the Main Distribution Panel (MDP) with a four conductor power cable supplied by the Installer. Size to be determined by Installer.	

Table 5 **Power Requirements**

3-3 EQUIPMENT CLEARANCE REQUIREMENTS

It is necessary to maintain proper distances at the front and back of the TGWC cabinet in order to have a good service area. The clearances shown in Illustration 2 include the area needed to slide out the trays containing the pump and compressor sections and to service them from the side.

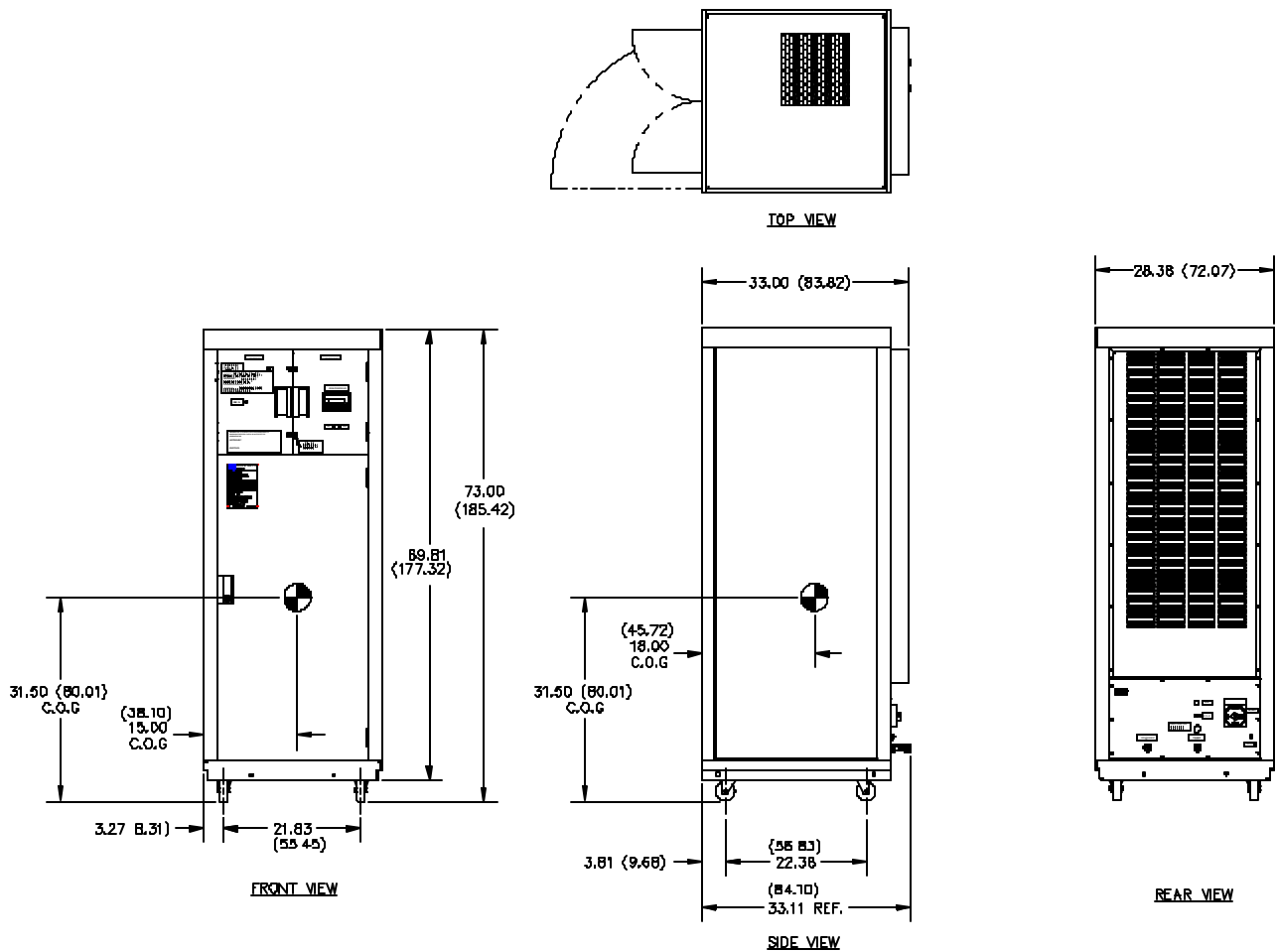


NOTE: The unit has two (2) fixed casters located in the rear of the unit and two (2) swivel casters located in the front of the unit.

Illustration 2 **NECESSARY CLEARANCES FOR TGWC**

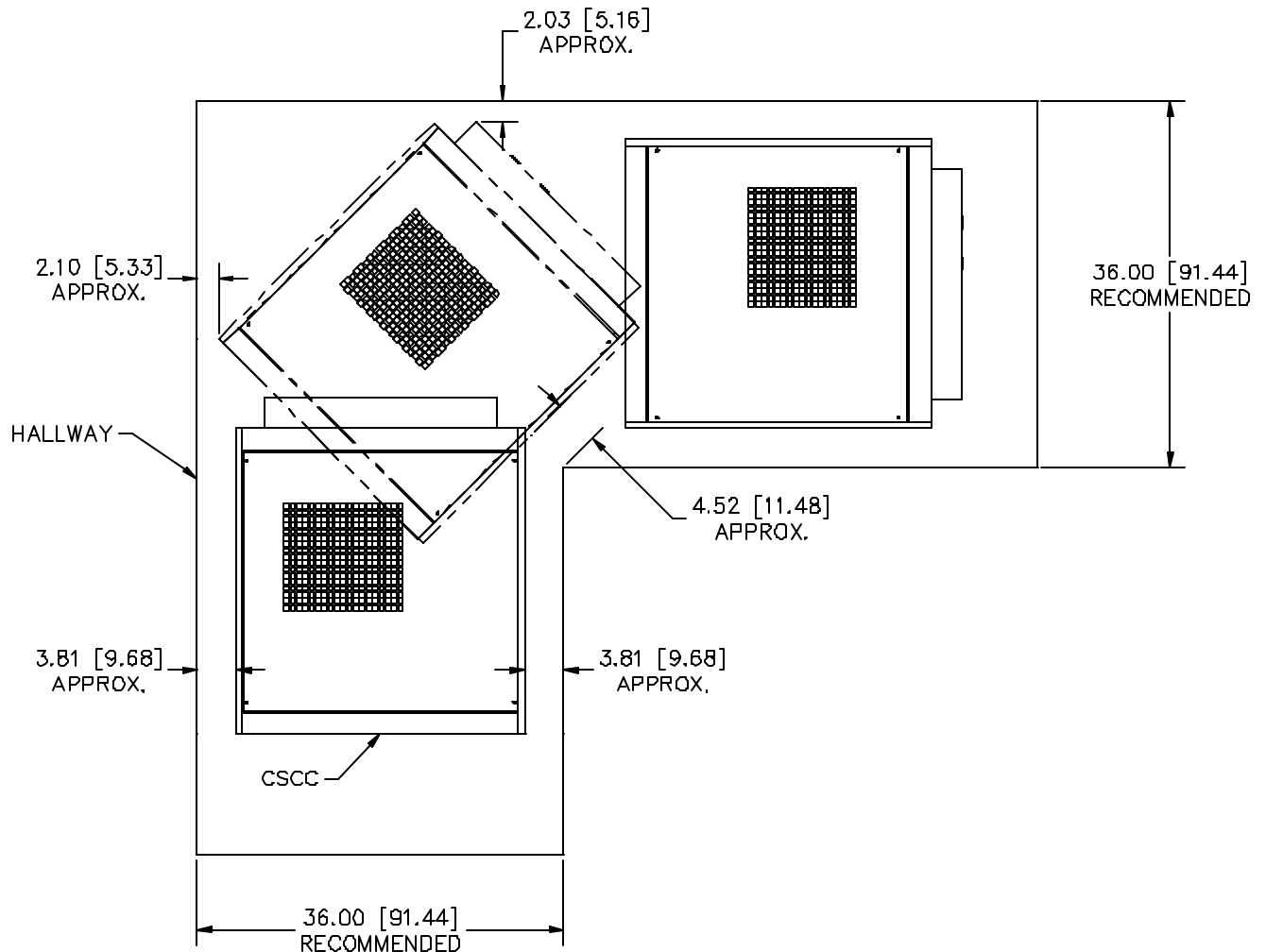
3-4 COMPONENT DIMENSIONS

Component	Height	Width	Depth
TGWC	73 inches (185 cm)	28.38 inches (72.10 cm)	33 inches (83.8 cm)

Table 6 **Uncrated Dimensions**Illustration 3 **TGWC OVERALL DIMENSIONS**

3-5 MINIMUM DOOR/HALLWAY SIZES

Component	Hallway / Door Height	Hallway Width Note	Doorway Width
TGWC	83 inches 210.82 cm	36 inches 91.44 cm	36 inches 91.44 cm
NOTE: Minimum width for 90° turns			

Table 7 **Door/Hallway Sizes**Illustration 4 **MINIMUM HALLWAY DIMENSION FOR TURNS ([] = cm)**

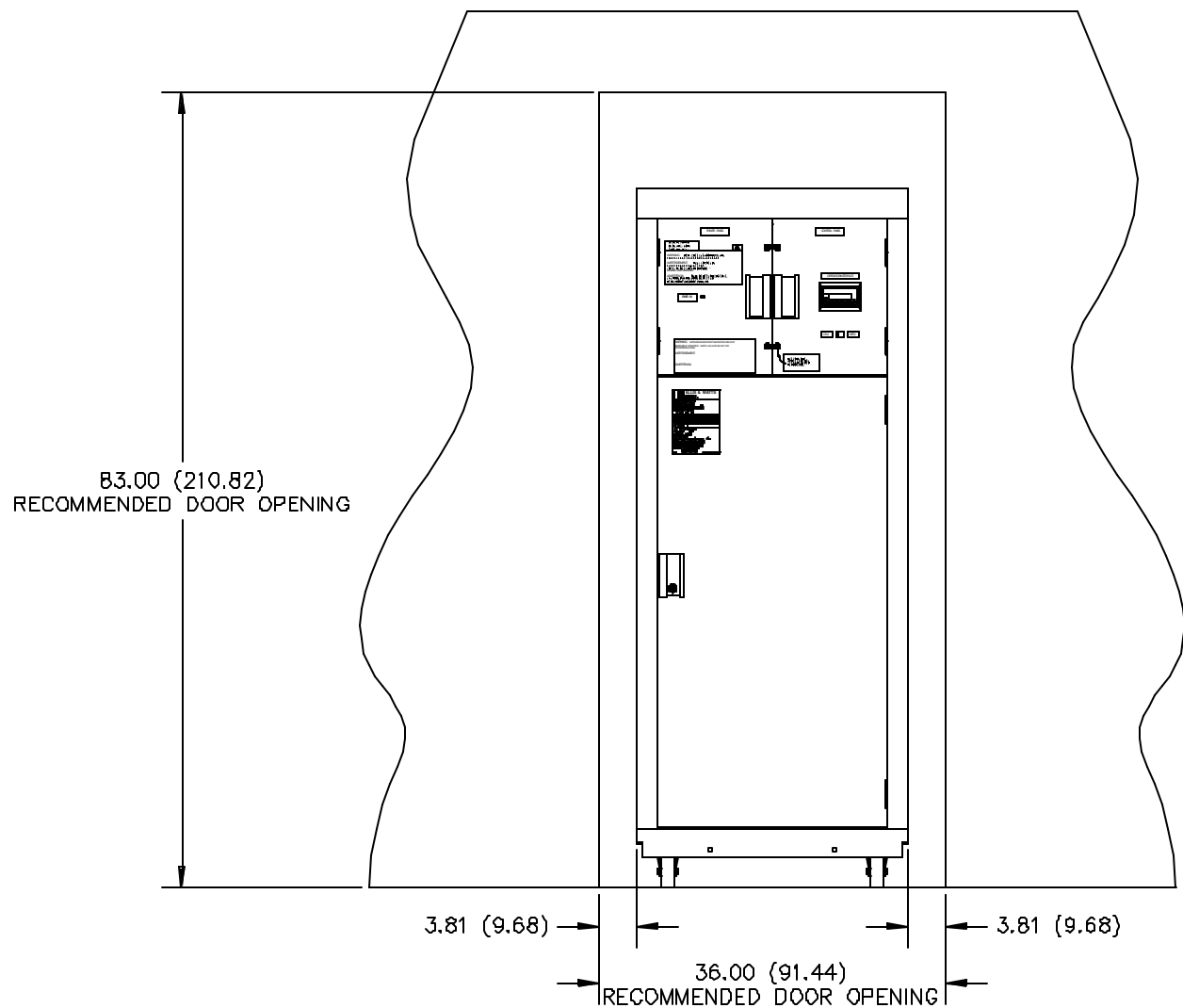


Illustration 5 **MINIMUM DOORWAY DIMENSIONS ([] = cm)**

3-6 DOORWAY THRESHOLD CONSIDERATION

The clearance from the bottom of the TGWC cabinet to the floor is approximately 3.36 inches (8.53 cm) therefore doorway threshold should be as smooth as possible or special considerations have to be made to move cabinet over a threshold height of 3 inches (7.62 cm)

3-7 FLOOR LOADING AND WEIGHTS

EQUIPMENT	INSTALLED WEIGHT
TGWC	780 lbs.(354.5 Kg)
NOTE Supported with 2 swiveling and 2 rigid casters	

Table 8 **Floor Loading Weights**

3-8 ANCHORING AND SEISMIC CONSIDERATIONS

The TGWC cabinet has tie down brackets included in the ship loose parts, which should be used to aid in preventing the cabinet from shifting during a seismic event. Site architects and Engineers need to take into consideration the raised floor-framing layout in order to have substantial support for the tie down brackets.

Sites requiring seismic anchoring by code should have the site architect and engineer review the response spectra and/or United Builders Code (UBC) for their location and then contact Ellis and Watts International for assistance in seismic planning of the site.

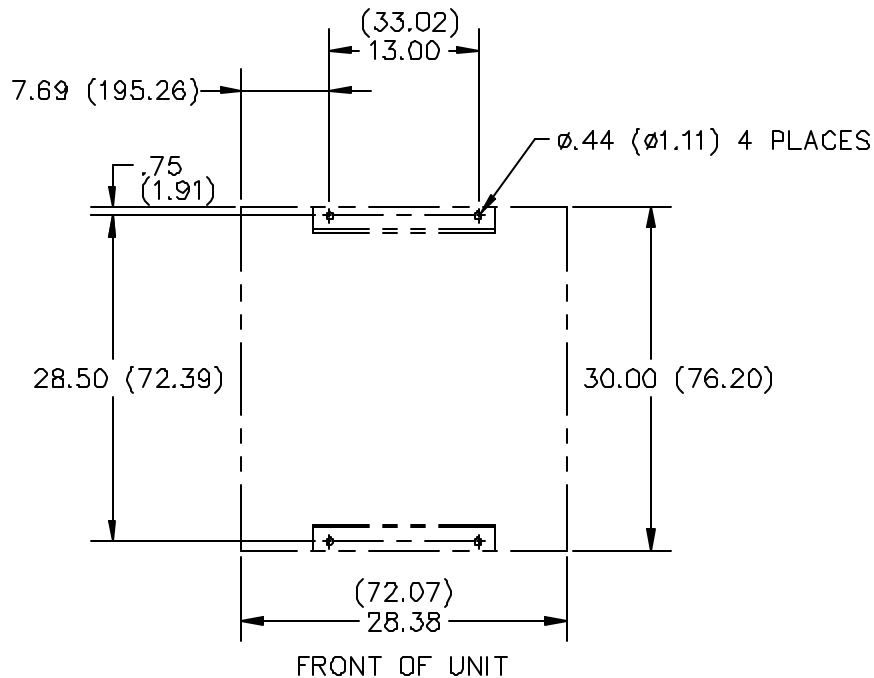


Illustration 6 **TGWC TIE DOWN BRACKET PLACEMENT**

4 UNCRATING AND INSTALLATION PROCEDURES

4-1 UNCRATING AND RECEIVING INSPECTION

4-1-1 TGWC International Crating

1. Using a Phillips screwdriver, unscrew the wood screws holding on the top of the crate.

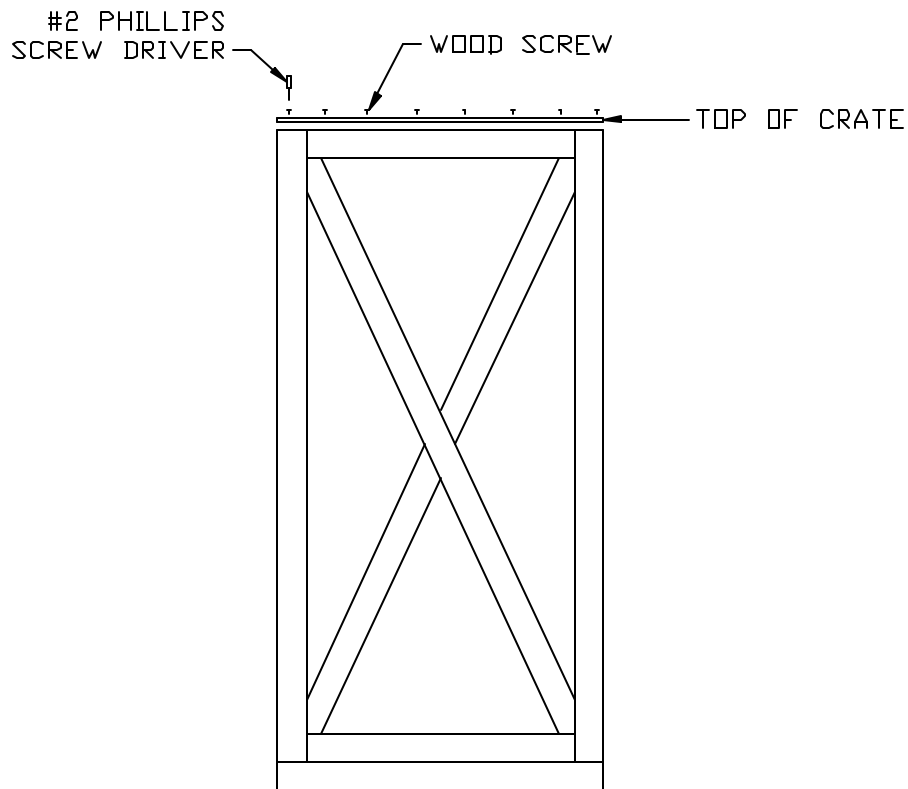


Illustration 7 **INTERNATIONAL SHIPPING CRATE FOR TGWC INDOOR UNIT**

2. Remove the top of the crate and place in an area clear of the uncrating area.
3. Using a pry bar, carefully remove the front, sides, and back of the crate and place in an area clear of the uncrating area.

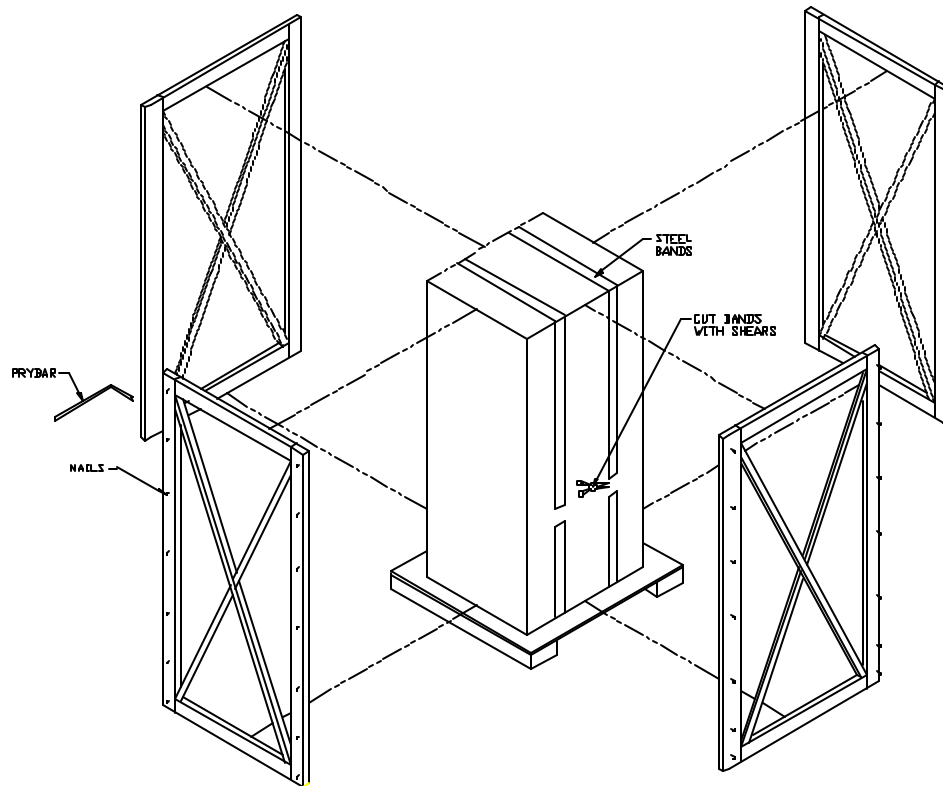


Illustration 8 **INTERNATIONAL UNCRATING**

CAUTION

BANDING STRAPS CAN BE UNDER HIGH TENSION AND CAN WHIP OUT WHEN CUT.

4. Remove banding by cutting with metal cutting hand shears.
5. CAREFULLY remove plastic
6. Inspect equipment for any damage. Contact Ellis & Watts and the shipping company as soon as possible and report any discovered damage.

4-1-2 TGWC Domestic Crating

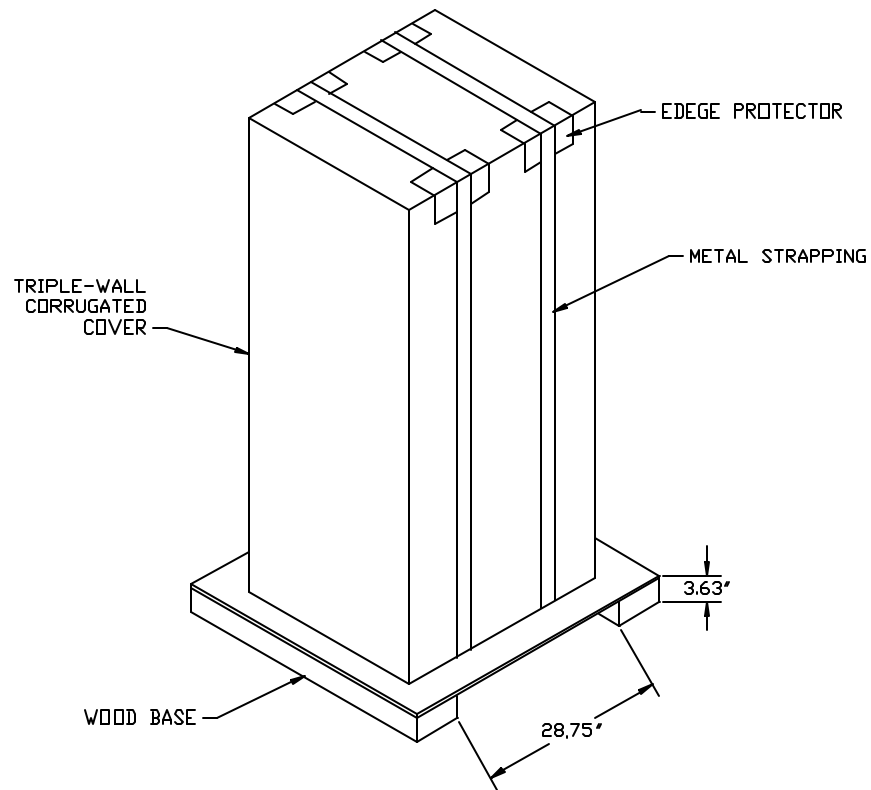


Illustration 9 **DOMESTIC SHIPPING CRATE FOR TGWC INDOOR UNIT**

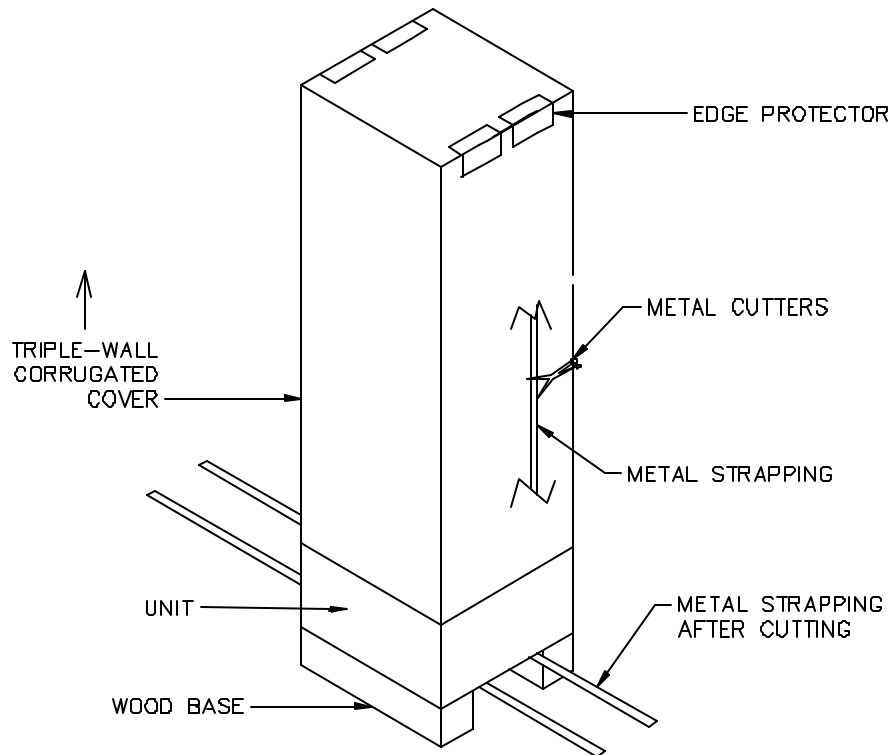


Illustration 10 **DOMESTIC UNCRATING**

CAUTION

BANDING STRAPS CAN BE UNDER HIGH TENSION AND CAN WHIP OUT WHEN CUT.

1. Remove banding by cutting with metal cutting hand shears.
2. Remove edge protectors from around top of unit.
3. Remove fasteners from along base of cardboard cover using pliers.
4. Remove cardboard cover from unit and discard.
5. Inspect unit for damage. Contact Ellis and Watts and the shipping company as soon as possible and report any discovered damage.
6. Using two eye and eye lifting straps approximately 2 inches (5.08 cm) wide and 18 feet (5.49 m) long, create a basket lift by pushing one end under unit (between board slats from front to back), and up opposite side until eyes are even.
7. Do this in two places, evenly spaced from, and as near as possible to sides of TGWC as fork spread will allow.

8. Using one 10 foot (3.05 m) stepladder on one end of the unit, hold up straps while bringing in forklift as near as possible to front or back of TGWC cabinet so lift forks can be inserted into strap eyes.

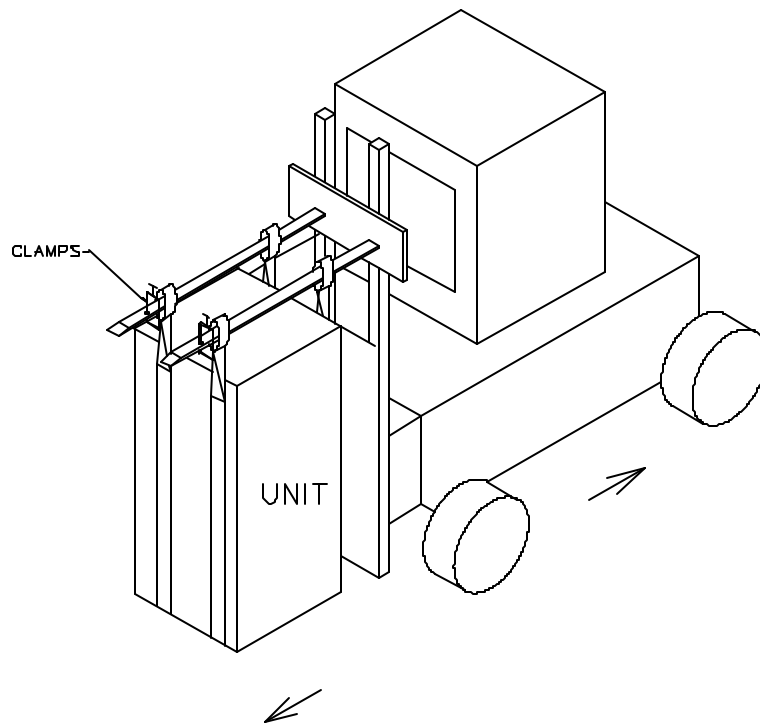


Illustration 11 **TGWC LIFTING METHOD**

9. After sliding lift forks through strap eyes, attach a C-clamp or similar clamping device, onto the end of the forks to prevent straps from sliding off the forks

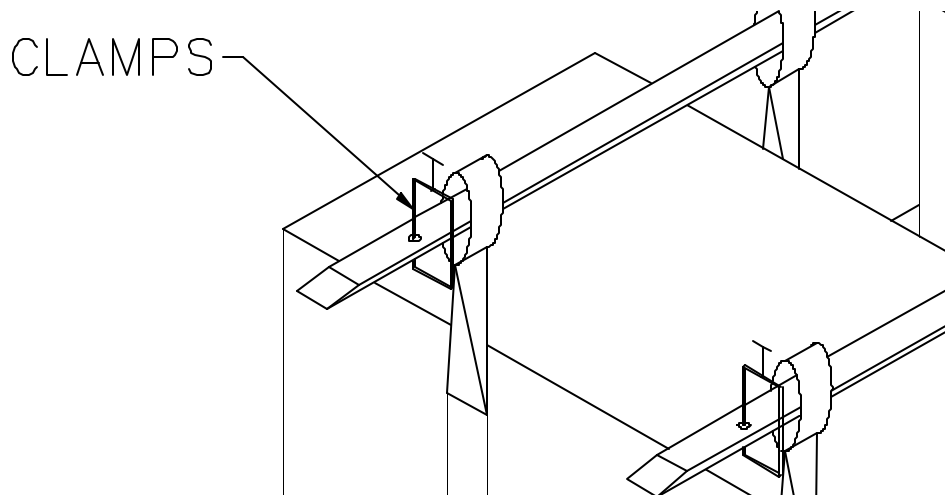


Illustration 12 **C-CLAMP PLACEMENT**

WARNING

STAND CLEAR DURING LIFTING OPERATIONS. OPERATOR ERROR AND/OR EQUIPMENT MALFUNCTION CAN CREATE OPPORTUNITIES FOR PERSONAL INJURY.

10. Gently lift TGWC cabinet until clear of shipping skid.
11. Remove skid, carefully move unit to smooth level area where TGWC cabinet can be moved on attached casters.
12. Remove slings from lifting forks and slide out from under TGWC cabinet.

The previous steps define the preferred method of removing the TGWC cabinet from the skid. Other methods of removing the TGWC cabinet from the shipping skid may be acceptable but could involve many more detailed steps. It is up to the judgment of the riggers and movers to determine the specific method of handling the TGWC cabinet.

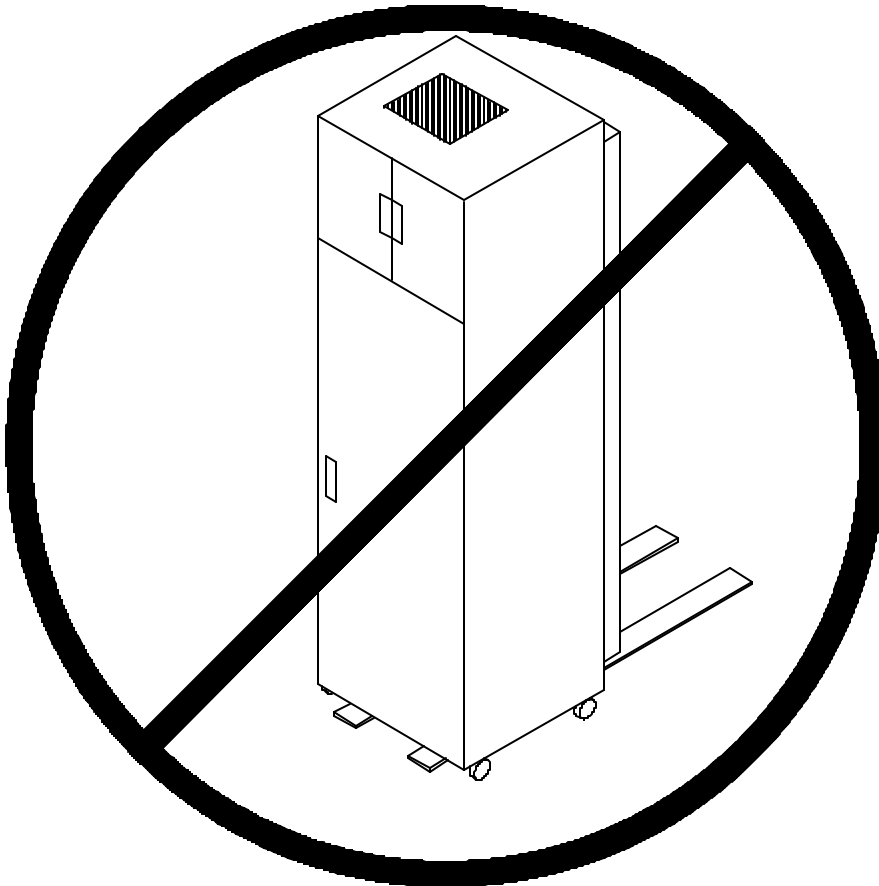


Illustration 13 UNACCEPTABLE ALTERNATIVE LIFTING METHOD

4-2 SHIP LOOSE ITEMS

The following tables list the ship loose items that are necessary for the proper installation of the Ellis and Watts equipment. These items were shipped inside the TGWC. The following tables can be used to verify the inventory of the shipped loose items. Also, the field installation drawings FI-0014, FI-0097, FI-0017, FI-0092 and FI0093 are included in the ship loose kits.

The TGWC is shipped with the following field installation kits.

No.	Part No.	Ship Loose Items Description	QTY EA
1	T2624-172A	Anti Tip Legs with Mounting Hardware	2
2	FI-0014	Field Installation Anti-Tip Legs	1

Table 9 **Anti Tip Leg Install Kit**

No.	Part No.	Ship Loose Items Description	QTY EA
1	SS627-1-A20	Fuse, Time Delay, 1.0 Amp 500 V	2
2	J1520-5050	Propylene Glycol, Premixed, 1 Gallon GE P/N 2297672	10
3	FI-0097	Field Installation Spare Parts Kit	1

Table 10 **Spare Parts Kit**

No.	Part No.	Ship Loose Items Description	QTY. EA
1	T2624-0201-21	Seismic Tie Down Bracket	2
2	50-19X-633	Screw, truss head Phillips 1/4-20 x 1 1/4 inch LG. SST	4
3	50-3E-34	Washer, flat, 1/4 inch SST	4
4	50-2Z-126	Nut, Lock 1/4-20 SST	4
5	FI-0017	Field Installation Seismic Tie Down Bracket	1

Table 11 **Seismic Tie Down Bracket Install Kit**

No.	Part No.	Ship Loose Items Description	QTY EA
1	T2626-142ZS	3/4 inch Quick Disconnect Nipple, 3/4 inch Female NPT	1
2	T2636-140ZS	3/4 inch Quick Disconnect Coupler, 3/4 inch Female NPT	1
3	J1476-206	3/4 inch Hose Barb, 3/4 inch male NPT	2
4	FI-0093	Field Installation Piping/Hose Connections	1

Table 12 **Piping/Hose Connection Install Kit**

NOTE: FI-0092 has no ship loose parts. It will be packed separately with the other ship loose kits.

5 EQUIPMENT INSTALLATION

5-1 SEISMIC TIE DOWN BRACKETS

Refer to drawing FI-0017 that comes with the ship loose parts, for detailed illustrations.

1. Locate-Tie Down Brackets in the ship loose kit FI 0017
2. Locate positions to mount brackets per Tie-Down Brackets Layout Illustration.

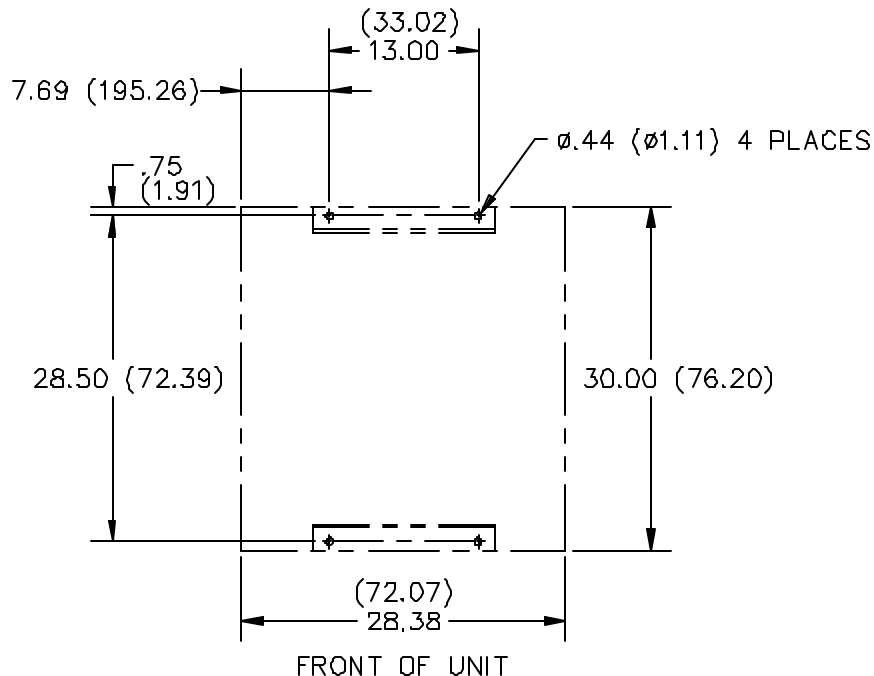
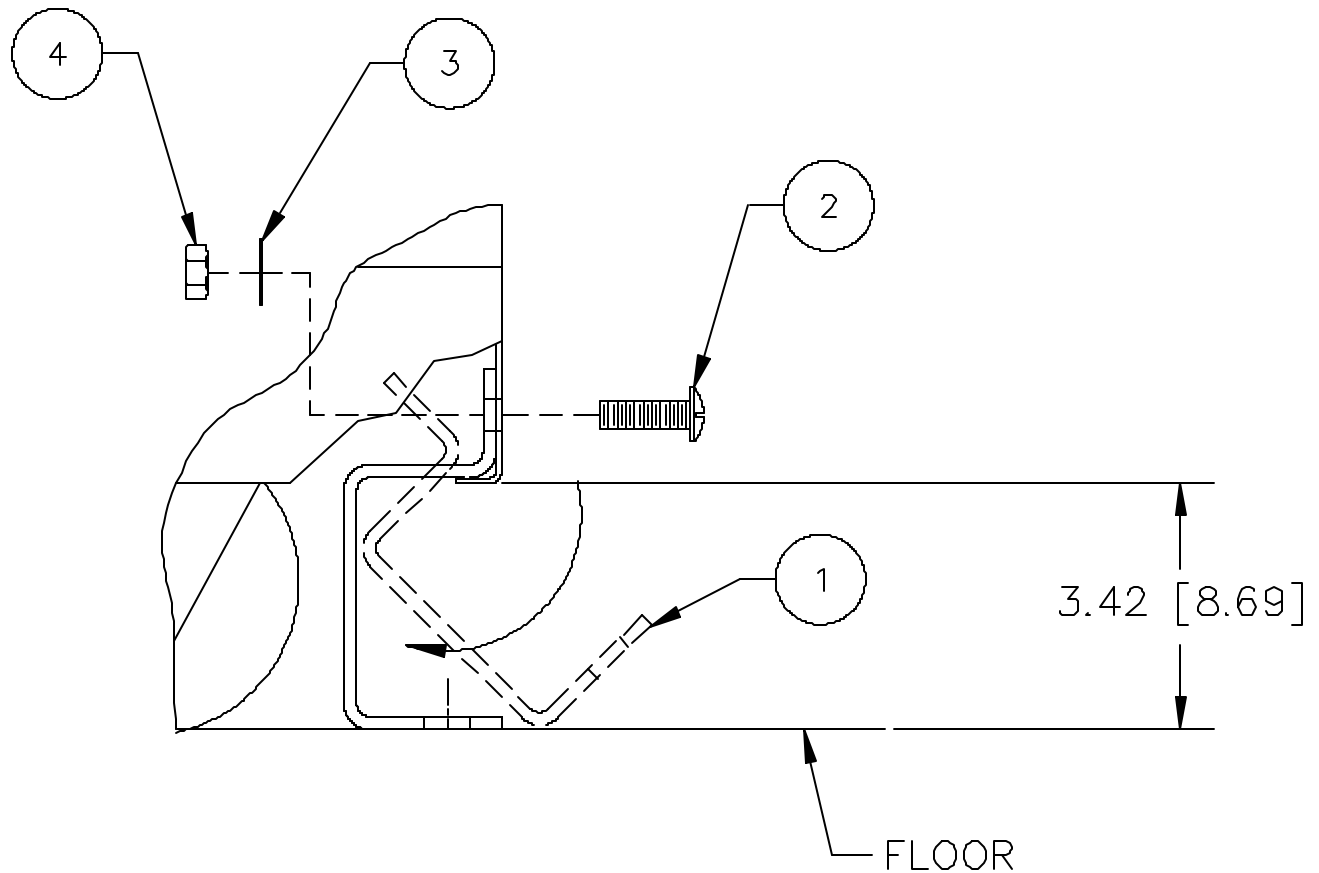


Illustration 14 **TIE-DOWN BRACKET LAYOUT PATTERN**

3. Locate mounting hardware in ship loose kit FI-0017. Refer to Table 11, items 1-3.
4. Align tie down bracket with the side of the frame as shown in Illustration 15 on next page.
5. Taking one $\frac{1}{4}$ -20 x 1 $\frac{1}{4}$ inch long truss head screw, slide one $\frac{1}{4}$ inch washer over screw and then push the screw through the hole in the side of the frame and through the hole in the tie down bracket. Slide one $\frac{1}{4}$ inch washer over screw and thread $\frac{1}{4}$ inch lock nut down until it is snug against the frame member and tighten approximately $\frac{1}{4}$ to $\frac{1}{2}$ turn more.

**Illustration 15 TIE DOWN MOUNTING BRACKET**

Refer to Table 11 items 1 to 4.

6. Repeat steps 3 to 5 for the other bracket.
7. Using brackets as templates, mark locations for floor mounting hardware.
8. Drill marked locations to the appropriate size for installer supplied hardware. Ellis & Watts recommends 3/8 inch hardware.
9. Determine torque requirements for floor mounting hardware and tighten to proper torque.

5-4 ANTI TIP LEG INSTALLATION

Refer to drawing FI-0014 for detailed illustrations.

NOTE: Anti-Tip legs are not needed when the seismic tie down brackets are used.

1. Locate Anti-Tip Legs in the ship loose kit FI-0014.
2. Align leg with threaded inserts located on the left and right side of the cabinet frame.
3. Attach each leg by threading two (2) 3/8-16 x 1 1/4 inch pan head bolts furnished with anti-tip legs into the inserts in the bottom of the cabinet frame.
4. Repeat with other anti-tip leg.
5. Adjust leg to achieve desired cabinet level by rotating bottom leveling glide clockwise to decrease height or counterclockwise to increase height

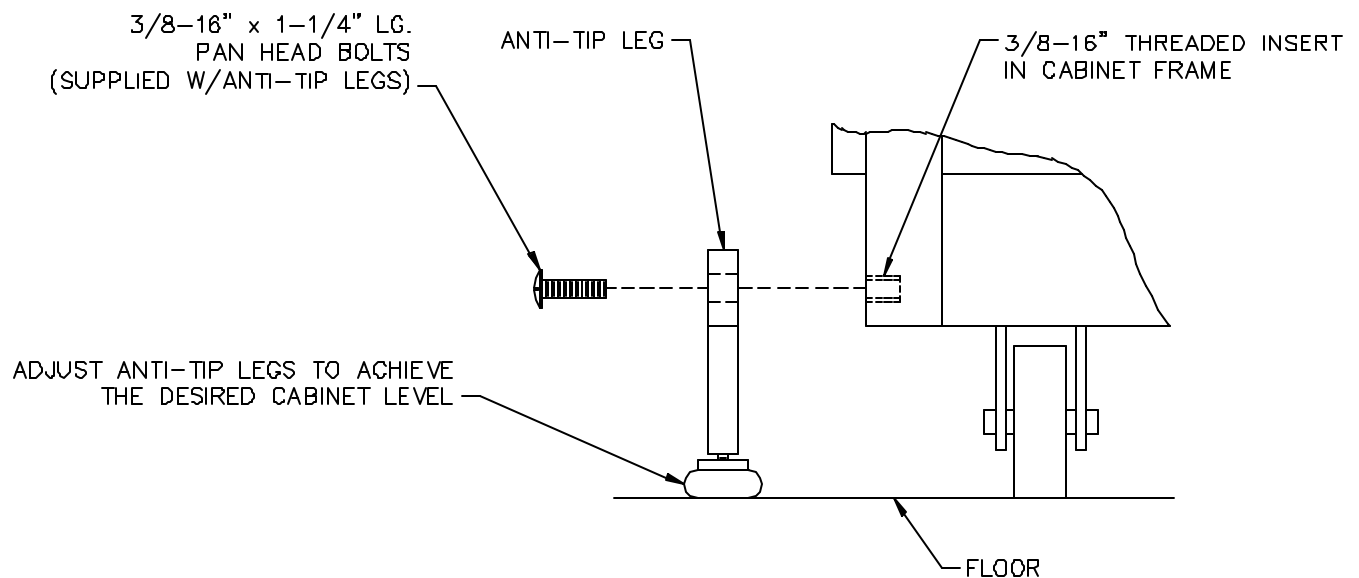


Illustration 16 ANTI TIP LEG INSTALLATION

5-5 ELECTRICAL INSTALLATION

Power Wiring

Connecting power to the TGWC should only be performed by trained electrical installation personnel that have a good knowledge of lock-out/tag-out procedures. Circuit and field wiring must be accomplished in accordance to the National Electrical code, ANSI/DEPA No. 70-1990 and or the Canadian Electrical Code, Part 1, CSA Standard C22.1-1990 and local laws and regulations as they exist.

Power Requirement

480 VAC $\pm$ 10% @60 Hz $\pm$ 3 Hz, 3 phase, WYE or Delta
or
400 VAC $\pm$ 10% @50 Hz $\pm$ 3 Hz, 3 Phase, WYE or Delta
Full load is 11.55 KVA.

Refer to drawing FI-0092 that comes with the ship loose parts, for detailed illustrations.

1. Locate the disconnect switch on the back lower right side of the TGWC.

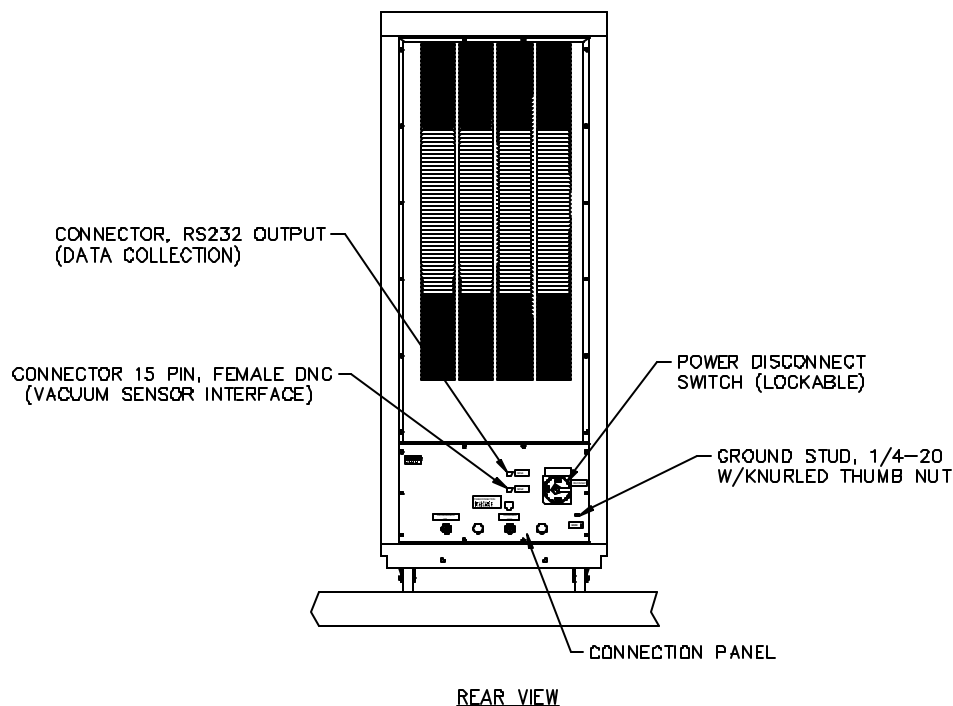


Illustration 17 DISCONNECT SWITCH LOCATION

2. Remove the connection panel by loosening the six screws securing the panel to the unit.

NOTE: Ellis & Watts recommends using a minimum conductor size of 8 AWG (10 mm²) based on the protective circuit device, Maximum size recommended is 4 AWG (25 mm²). Ellis & Watts recommends using 1 ½ inch (minimum) Liquidtight (or equivalent) conduit and fittings. The installer supplies wire, conduit, and fittings.

3. Remove the guard protectors from the back of the switch (See Illustration 18).
4. Route the four power conductors through the 1 7/8" hole provided on the panel.
5. Strip the wire and insert into the terminals. When the conductor is properly installed, there should be no bare wire exposed. Pay special attention to the phase rotation.

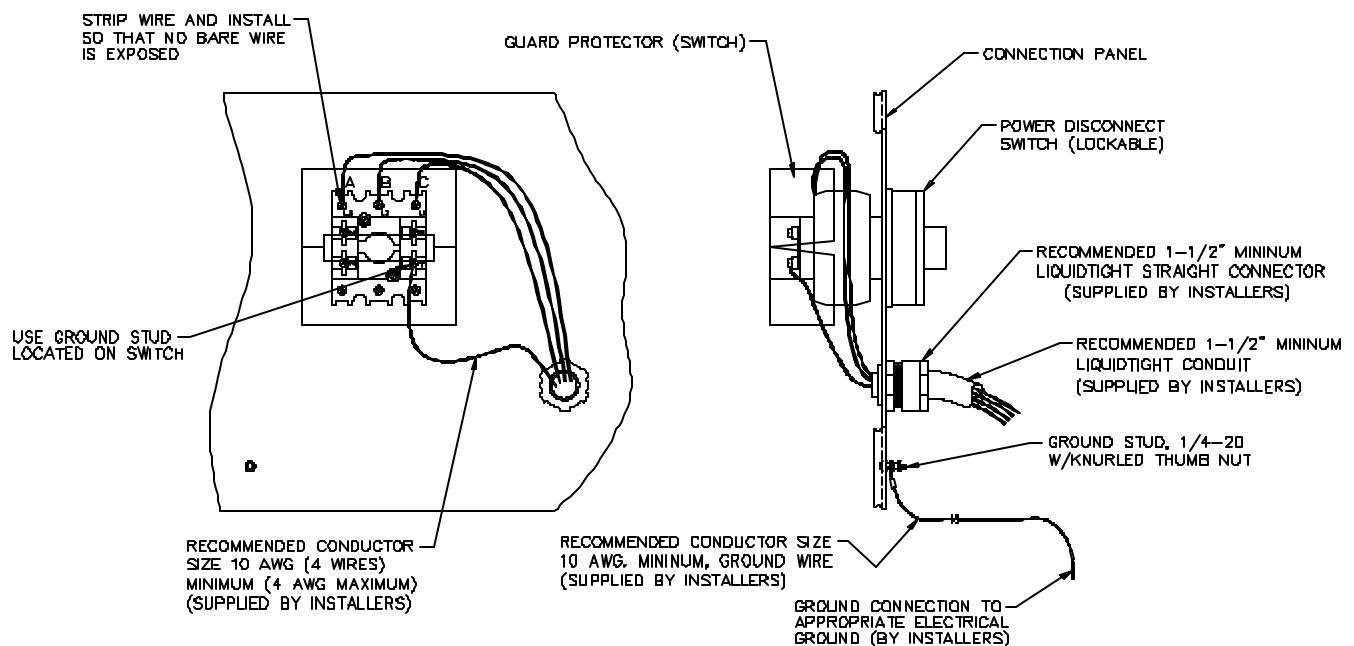


Illustration 18 **DISCONNECT SWITCH WIRING**

6. The ground stud provided on the terminal is for attaching the ground wire.
7. Reinstall the guard protectors and panel cover.

5-4 ASSEMBLING THE INTERCONNECTING HOSES

1. Verify the parts and quantities from the FI-0093.

The following supplies are supplied by the installer and are required for the proper installation of the interconnecting hoses.

- Light oil or soapy water
 - Loctite 567 or equivalent
 - Hose $\frac{3}{4}$ inch
 - $\frac{3}{4}$ to 3 inch Hose Clamps
2. Using a razor knife, cut hose supplied by customer to predetermined length.
 3. Lubricate the barb fitting and/or hose I.D. with a light oil or soapy water only. Do not use heavy oil or grease.

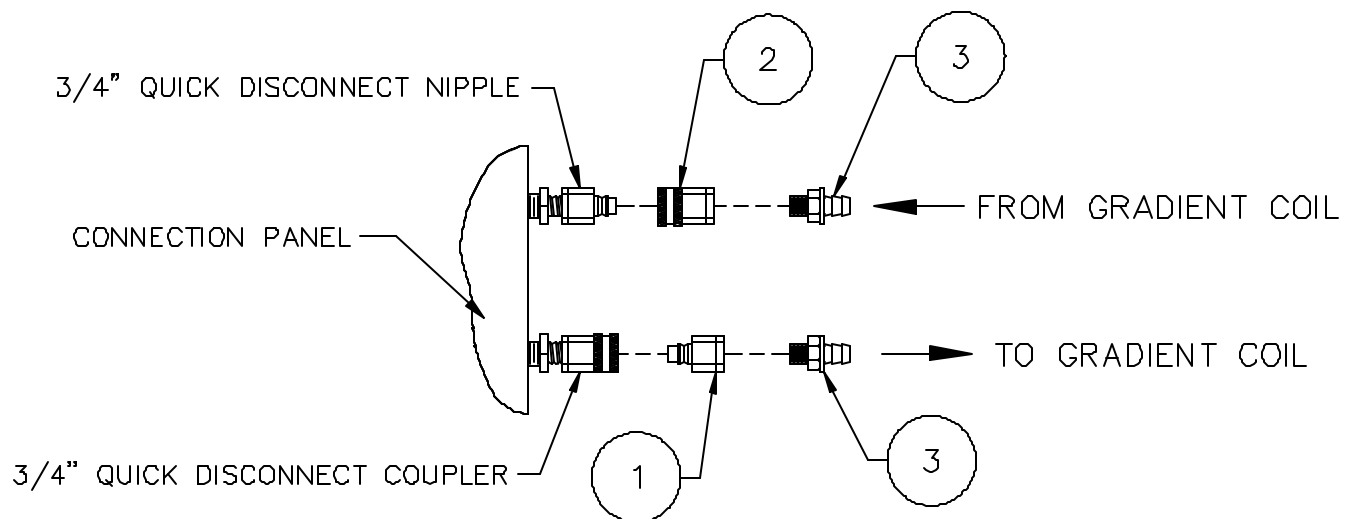


Illustration 19 QUICK DISCONNECT TO HOSE BARB ASSEMBLY

Refer to Table 12 items 1 to 3.

4. Insert the fitting into the hose until the barb is in the hose.
5. Place the end of the fitting against a wood block. Grip the hose firmly at approximately one inch (2.54 cm) from the end of the hose. Push with steady force until the end of the hose bottoms on the fitting and is covered by the yellow plastic cap.
6. Apply a generous amount of LocTite 567 or equivalent to the male pipe threads of each hose barb fitting.

7. TIGHTEN EACH THREADED CONNECTION SNUGLY TO PREVENT LEAKS.
8. Using hose clamps (GE supplied) attach each hose assembly to appropriate connection of gradient coil.

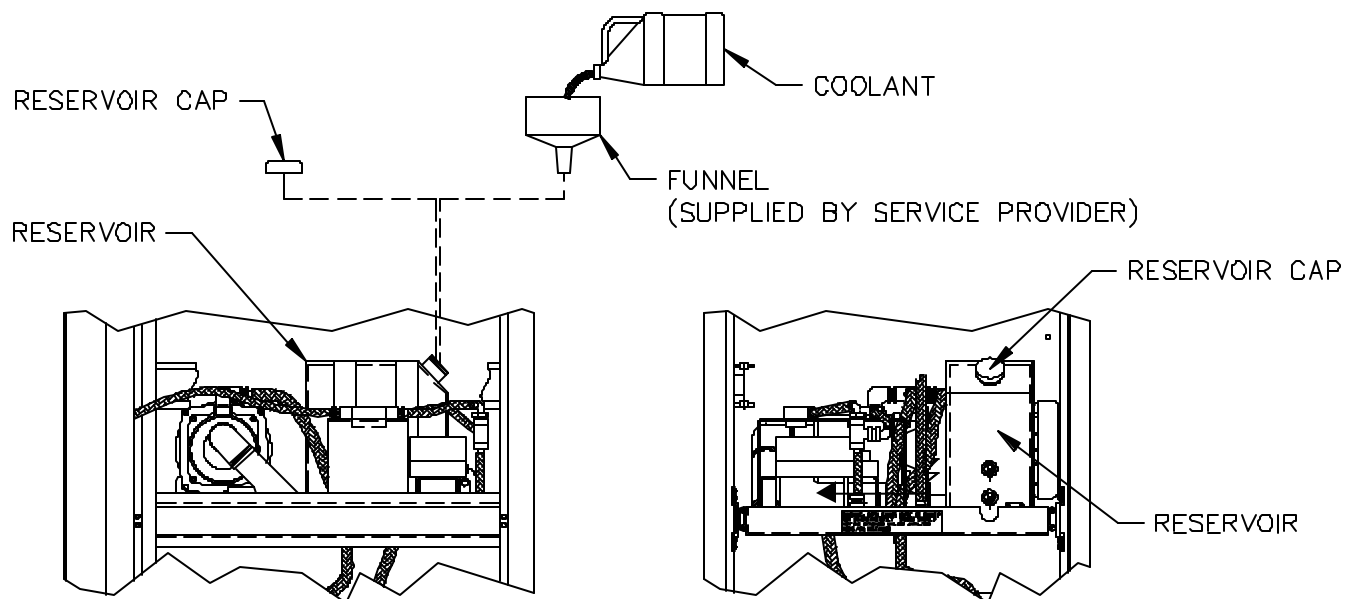
6 SYSTEM OPERATION

The following procedures are intended to give instruction on the TGWC system controls and routine operation. Initial start-up after installation will be by GE personnel. After installation, the system operation will be verified by an Ellis and Watts approved service provider. It is assumed at this point, that the TGWC and associated equipment have gone through initial startup previously and is ready for operation. If the unit has stopped due to a failure, refer to section 6-1-4.

6-1 OPERATING PROCEDURES

6-1-1 Starting Up The TGWC

1. Verify that all fluid lines are connected and not leaking. The fluid lines are located on the rear of the unit.
2. Check the coolant level in the reservoir. Verify that the coolant level is between the add and full labels. If it is not refer to Section 6-1-4. The reservoir is located on the middle shelf inside the front door (See Illustration 20).

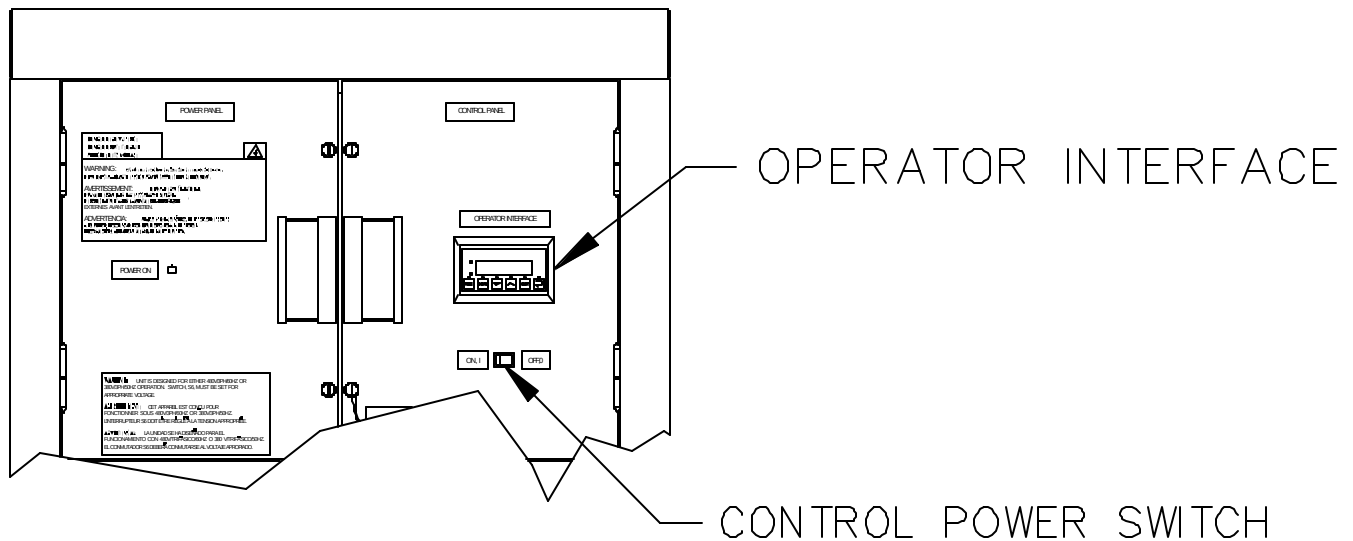


RESERVOIR COOLANT FILLING ILLUSTRATION

Illustration 20 **RESERVOIR FILL**

CAUTION

3. **DURING PERIODS WHERE THE COMPRESSOR CRANKCASE HEATERS ARE WITHOUT POWER, LIQUID REFRIGERANT CAN ACCUMULATE IN THE COMPRESSOR. RESTARTING THE COMPRESSOR WITHOUT ALLOWING SUFFICIENT TIME FOR THE CRANKCASE HEATERS TO VAPORIZE THE LIQUID REFRIGERANT CAN CAUSE SERIOUS INTERNAL DAMAGE TO THE COMPRESSOR. WAIT A MINIMUM OF TWO HOURS AFTER APPLYING POWER BEFORE CONTINUING.**
4. The Control Power Switch is located on the top front panel (see Illustration 21). Verify toggle switch is switched off. If the main disconnect switch is OFF, set to ON by rotating red handle clockwise until handle is in vertical position. The main disconnect switch is located on the back lower right side of the unit (see Illustration 17).
5. Place the Control Power Switch to the ON Position. TGWC should start after 5 seconds and be fully functional after 30 seconds. If the TGWC shuts down after 30 seconds, refer to section 6-1-4.

Illustration 21 **CONTROL PANEL**

6. After 30 minutes of operation verify that the unit is maintaining proper temperatures and flows as shown in table 13.

Equipment	Coolant Flow	Coolant Temperature
TGWC	6.0 GPM +/- 10%	20° +/- 1C

Table 13 **TGWC Temperature And Flow Tolerances**

6-1-2 Shutting Down The TGWC.

1. Verify that the red alarm LED on the operator interface is not on. If the LED is glowing, check the alarm section of the operator interface for any faults and log them before proceeding to prevent the loss of fault information. Refer to section 6-1-4 for further instructions.
2. Set the Control Power Switch to OFF.

6-1-3 Maintenance

An approved service provider should conduct routine maintenance every 4 months. This maintenance consists of:

Verifying proper flows and temperatures per table 13 and logging data.

Change coolant filter.

Leak test refrigerant system.

Check electrical control panel for burned or frayed wiring and loose connections.

Checking all coolant hoses for obvious leaks and for worn areas.

Verifying proper operation of refrigeration system.

Check condenser coils for dirt or damage.

Contact GE for current schedule or approved service provider.

6-1-4 General Troubleshooting

The TGWC chiller system is designed to be monitored by the GE Service Center at all times. Some occasions arise where the operator can assist in troubleshooting. In all cases the operator will be **most useful** by checking for and logging any faults on the operator interface and noting any abnormal conditions in the equipment room where the TGWC is located. Refer to sections 6-2-1 through 6-2-3 for instruction on how to locate faults on operator interface screens.

6-2 PLC OPERATION

6-2-1 Operator Interface Display Description:

The operator interface is located on the front panel of the TGWC and displays the model number of the unit, the software revision of the programmed PLC, parameter set points, parameter data, parameter alarm status, and system run time. The interface also allows the user to change the operating set points for the fluid coolant circuit. The upper and lower alarm limits for the various parameters are preset by the manufacturer and are not accessible by service. See Illustration 22 for an example of the layout of the display, user buttons, power switch and indicator lights.

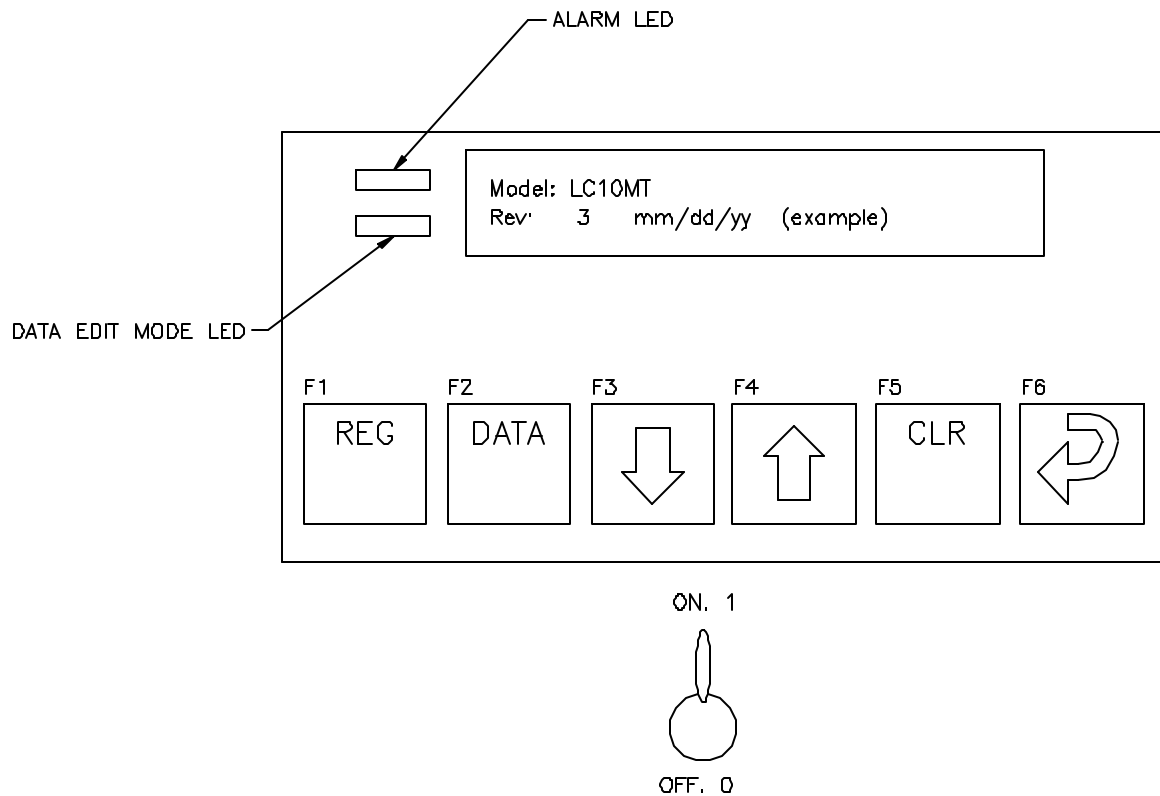


Illustration 22 **OPERATOR INTERFACE**

6-2-2 Button Description

Button F1 Toggles the display between Menu 1 and Menu 2.

Button F2 Puts the unit into data edit mode for changing the operating set point for supply coolant temperature. The green led will light when the unit is in data edit mode.

Button F3 Menu List Display: Steps through the menu list in the forward direction. Change Set Point: Increases the value shown on the display in 0.1 unit increments.

Button F4 Menu List Display: Steps through the menu list in the reverse direction. Change Set Point: Decreases the value shown on the display in 0.1 unit increments.

Button F5 To extinguish the red alarm status indicator. Note: The operating parameter must be in a normal or no fault mode in order to turn off the red alarm status indicator.

Button F6 Stores new set point changes and exits the data edit mode.

6-2-3 Menu Lists

1. Press the F1 Button to change between Menu 1 and Menu 2.
2. Refer to Table 14, Menu 1 – System Parameter Data, to view the listing of the following types of items: Model Number and Software revision of the unit, operating set points for specific parameters, current parameter data measurement, and total run hours. The first line item on Menu 1 will show the Model Number of the Unit.
3. Refer to Table 15, Menu 2 – System Operating Status, to view the listing of the alarm or failure conditions. The first line item on Menu 2 will show “Supply Fluid Temperature”.

6-2-4 Changing The Set Points

The coolant temperature set points are preset at Ellis and Watts per G.E. specifications. There is a 2°C allowable variance, settable by the operator. If there is a need to change the set point value, then refer to the following steps to perform the change.

1. Press F3 to step forward through the menu until the display shows the coolant temp set point or the TGWC air temp set point.
2. Press F2 to enter the data edit mode.
3. Press either the F3 or F4 buttons to change the set point value.
4. Press F6 to store the set point values and exit the data edit mode.

Example of Model Number and Revision**Model Number: LC10MT****Rev.: # Mo/Dy/Yr**

Parameter	Range Value / Unit of Measure (see Note 1)
Supply Temp:	0 to 29.4 Degrees Centigrade
Return Temp:	0 to 50 Degrees Centigrade
Temp Set Point:	17 to 24 Degrees Centigrade (Factory set 20°C)
Gradient Flow:	0 to 99.99 GPM
TGWC 3 way valve % open:	0 to 100%
Total Run Hours:	0 to 9999 Hours
NOTE 1: Unit of measure is set at the factory.	
NOTE 2: After total run hours reaches 9999 hours, the unit resets to zero.	

Table 14 **Menu 1 – System Parameter Data**

Parameter	Display of Operating Mode		
Supply Temp:	Low	Normal	High
Coolant Level:	Low	Normal	
Gradient Flow:	Failure	Low	Normal
Suction Pressure	Fault	No Fault	
Discharge Pressure	Fault	No Fault	
Compressor / Phase	Fault	No Fault	
Compressor Cycling:	Normal	- - - - (See Note 1)	
NOTE 1: Each dashed line represents an attempt by the PLC to restart the chiller. After 5 attempts the TGWC will shut down.			

Table 15 **Menu 2 – System Operating Status**

TERMS & ACRONYMS

GEMS	General Electric Medical Systems
TGWC	Twin Gradient Water Cooler
MDP	(Main Distribution Panel)
QD	Quick disconnect coupler
RGA	Returned Goods Authorization
EPA	Environmental Protection Agency
MRI	Magnetic Resonance Imaging
UBC	United Builders Code
SST	Stainless Steel
NPTF	National Pipe Thread, Female
DB	Distribution Block
PLC	Programmable Logic Control
Liquidtite	Trade name for water tight flexible conduit
LocTite	Brand name for adhesives, sealants, and threadlockers

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MATERIAL SAFETY DATA SHEETS

Forane ® R - 134a

Tetrafluoroethane (HFC - 134a)

Dowfrost ® Heat Transfer Fluid

Propylene glycol

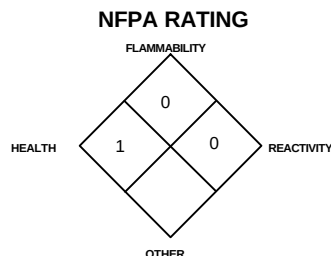
Dipotassium Phosphate

Deionized Water



MATERIAL SAFETY DATA SHEET

Prepared to U.S. OSHA, CMA, ANSI and Canadian WHMIS Standards



PART I What is the material and what do I need to know in an emergency?

1. PRODUCT IDENTIFICATION

CHEMICAL NAME; CLASS: **TETRAFLUOROETHANE - C₂H₂F₄**
HALOCARBON 134a
Document Number: 1055

PRODUCT USE: For general analytical/synthetic chemical uses.

SUPPLIER/MANUFACTURER'S NAME: AIRGAS INC.
ADDRESS: 259 Radnor-Chester Road
Suite 100
Radnor, PA 19087-5240

BUSINESS PHONE: 1-610-687-5253
EMERGENCY PHONE: CHEMTREC: 1-800-424-9300
International: 202-483-7616

DATE OF PREPARATION: May 22, 1996
SECOND REVISION: January 30, 1998

2. COMPOSITION and INFORMATION ON INGREDIENTS

CHEMICAL NAME	CAS #	mole %	EXPOSURE LIMITS IN AIR					
			ACGIH		OSHA		IDLH ppm	OTHER
			TLV ppm	STEL ppm	PEL ppm	STEL ppm		
Tetrafluoroethane	811-97-2	100	There are no specific exposure limits for Tetrafluoroethane. Tetrafluoroethane is an asphyxiant. Oxygen levels should be maintained above 19.5%. Note: DuPont has a recommended exposure level of 1000 ppm.					DFG MAK: 1000 ppm

NE = Not Established

C = Ceiling Limit

See Section 16 for Definitions of Terms Used

NOTE: All WHMIS required information is included. It is located in appropriate sections based on the ANSI Z400.1-1993 format.

3. HAZARD IDENTIFICATION

EMERGENCY OVERVIEW: Tetrafluoroethane is an odorless, colorless non-flammable, liquefied gas. Tetrafluoroethane can cause central nervous system depression after inhalation exposures. Symptoms of such overexposure can include drowsiness, fatigue, and weakness. At high concentrations, the gas can act as an asphyxiant, by displacing oxygen; therefore, exposure to high concentrations of this gas can be fatal. Frostbite can be caused by contact with rapidly expanding gases or the liquefied gas. This gas is not flammable or reactive in normal emergency response situations. However, if involved in a fire, Tetrafluoroethane can decompose to produce toxic gases (e.g., hydrogen fluoride, phosgene).

SYMPTOMS OF OVEREXPOSURE BY ROUTE OF EXPOSURE: The most significant route of overexposure for Tetrafluoroethane is by inhalation. The following paragraphs describe symptoms of exposure by route of exposure.

INHALATION: Exposures to high concentrations of this gas may cause central nervous system depression and cause sensitization of the heart to epinephrine. Effects of such overexposure can include light-headedness, giddiness, shortness of breath and in extreme cases, irregular heartbeats, cardiac arrest, and death.

High concentrations of this gas can cause an oxygen-deficient environment. Individuals breathing such an atmosphere may experience symptoms which include headaches, ringing in ears, dizziness, drowsiness, unconsciousness, nausea, vomiting, and depression of all the senses. The skin of a victim of overexposure may have a blue color. Under some circumstances of overexposure, death may occur. The effects associated with various levels of oxygen are as follows:

CONCENTRATION

12-16% Oxygen:

10-14% Oxygen:

6-10% Oxygen:

Below 6%:

SYMPTOMS OF EXPOSURE

Breathing and pulse rate increased, muscular coordination slightly disturbed. Emotional upset, abnormal fatigue, disturbed respiration.

Nausea and vomiting, collapse or loss of consciousness.

Convulsive movements, possible respiratory collapse, and death.

OTHER POTENTIAL HEALTH EFFECTS: Contact with liquid or rapidly expanding gases (which are released under high pressure) may cause frostbite. Symptoms of frostbite include change in skin color to white or grayish-yellow. The pain after such contact can quickly subside. Contact with the skin or eyes may cause slight irritation.

HEALTH EFFECTS OR RISKS FROM EXPOSURE: An Explanation in **Lay Terms**. Overexposure to Tetrafluoroethane may cause the following health effects:

ACUTE: The most significant hazard associated with Tetrafluoroethane is inhalation of high concentrations of Tetrafluoroethane. Such overexposure can cause oxygen deficiency. Symptoms of such exposures include respiratory difficulty, ringing in ears, headaches, dizziness, indigestion, nausea, and possible death. Contact with liquid or rapidly expanding gases may cause frostbite.

CHRONIC: There are currently no known adverse health effects on humans associated with chronic exposure to Tetrafluoroethane.

TARGET ORGANS: Respiratory system, cardio-vascular system.

HAZARDOUS MATERIAL INFORMATION SYSTEM			
HEALTH		(BLUE)	1
FLAMMABILITY		(RED)	0
REACTIVITY		(YELLOW)	0
PROTECTIVE EQUIPMENT			B
EYES	RESPIRATORY	HANDS	BODY
	See Section 8		See Section 8
For routine industrial applications			

See Section 16 for Definition of Ratings

PART II *What should I do if a hazardous situation occurs?*

4. FIRST-AID MEASURES

RESCUERS SHOULD NOT ATTEMPT TO RETRIEVE VICTIMS OF EXPOSURE TO TETRAFLUOROETHANE WITHOUT ADEQUATE PERSONAL PROTECTIVE EQUIPMENT. At a minimum, Self-Contained Breathing Apparatus and Personal Protective equipment should be worn.

4. FIRST-AID MEASURES (Continued)

Remove victim(s) to fresh air, as quickly as possible. Trained personnel should administer supplemental oxygen and/or cardio-pulmonary resuscitation, if necessary. Only trained personnel should administer supplemental oxygen.

In case of frostbite, place the frostbitten part in warm water. DO NOT USE HOT WATER. If warm water is not available, or is impractical to use, wrap the affected parts gently in blankets. Alternatively, if the fingers or hands are frostbitten, place the affected area in the armpit. Encourage victim to gently exercise the affected part while being warmed. Seek immediate medical attention.

Victim(s) must be taken for medical attention. Rescuers should be taken for medical attention, if necessary. Take copy of label and MSDS to physician or other health professional with victim(s).

5. FIRE-FIGHTING MEASURES

FLASH POINT: Not applicable.

AUTOIGNITION TEMPERATURE: Not applicable.

FLAMMABLE LIMITS (in air by volume, %):

Lower (LEL): Not applicable.

Upper (UEL): Not applicable.

FIRE EXTINGUISHING MATERIALS: Non-flammable gas. Use extinguishing media appropriate for surrounding fire.

Water Spray: YES Carbon Dioxide: YES

Dry Chemical: YES Halon: YES.

Foam: YES Other: Any "ABC" Class

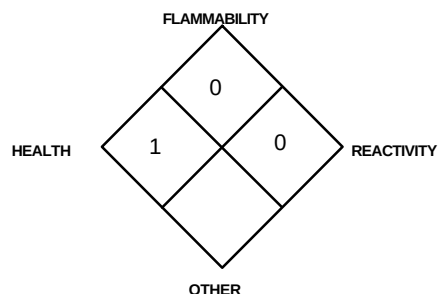
UNUSUAL FIRE AND EXPLOSION HAZARDS: Although Tetrafluoroethane is a non-flammable gas, it can present some health hazards to firefighters. When involved in a fire, this material may ignite and produce toxic gases (e.g., hydrogen fluoride, and carbonyl fluoride). Containers, when involved in fire, may rupture or burst in the heat of the fire.

Explosion Sensitivity to Mechanical Impact: Not sensitive.

Explosion Sensitivity to Static Discharge: Not sensitive.

SPECIAL FIRE-FIGHTING PROCEDURES: Structural firefighters must wear Self-Contained Breathing Apparatus and full protective equipment. In the event of fire, cool containers of Tetrafluoroethane with water to prevent failure. Use a water spray or fog to reduce or direct vapors. If unruptured cylinders are exposed to heat, the cylinder may rupture or burst and release the contents. It may be prudent to remove potentially heat-exposed cylinders from the area surrounding a fire, if it is safe for fire-fighters to do so.

NFPA RATING



See Section 16 for Definition of Ratings

Tetrafluoroethane does not burn; however,

6. ACCIDENTAL RELEASE MEASURES

SPILL AND LEAK RESPONSE: Uncontrolled releases should be responded to by trained personnel using pre-planned procedures. Proper protective equipment should be used. In case of an accidental release, evacuate all personnel upwind and away from affected area, protect people, and respond with trained personnel.

Minimum Personal Protective Equipment should be **Level B: mechanically-resistant gloves and Self-Contained Breathing Apparatus**. Allow the gas to dissipate. Monitor the surrounding area for oxygen. The atmosphere must have at least 19.5 percent oxygen before personnel can be allowed in the area without Self-Contained Breathing Apparatus. Attempt to close the main source valve prior to entering the area. If this does not stop the release (or if it is not possible to reach the valve), allow the gas to release in-place or remove it to a safe area and allow the gas to be released there.

PART III *How can I prevent hazardous situations from occurring?*

7. HANDLING and STORAGE

WORK PRACTICES AND HYGIENE PRACTICES: As with all chemicals, avoid getting Tetrafluoroethane IN YOU. Do not eat or drink while handling chemicals. Be aware of any signs of dizziness or fatigue; exposures to fatal concentrations of Tetrafluoroethane could occur without any significant warning symptoms.

STORAGE AND HANDLING PRACTICES: Cylinders should be stored in dry, well-ventilated areas away from sources of heat. Compressed gases can present significant safety hazards. Store containers away from heavily trafficked areas and emergency exits.

7. HANDLING and STORAGE (Continued)

SPECIAL PRECAUTIONS FOR HANDLING GAS CYLINDERS (continued): Post "No Smoking or Open Flames" signs in storage or use areas. Do not use Tetrafluoroethane with or near incompatible chemicals, such as the following: sodium, potassium, calcium, zinc, and magnesium, powdered aluminum. Acceptable materials for construction for equipment used in the handling of Tetrafluoroethane include most commonly used metals (steel, cast iron, brass, copper, tin, lead).

Protect cylinders against physical damage. Store in cool, dry, well-ventilated area, away from sources of heat, ignition and direct sunlight. Do not allow area where cylinders are stored to exceed 52°C (125°F). Use a check valve or trap in the discharge line to prevent hazardous backflow. Post "No Smoking or Open Flame" signs in storage and use areas. Cylinders should be stored upright and be firmly secured to prevent falling or being knocked over. Cylinders can be stored in the open, but in such cases, should be protected against extremes of weather and from the dampness of the ground to prevent rusting. Never tamper with pressure relief devices in valves and cylinders. The following rules are applicable to work situations in which cylinders are being used:

Before Use: Move cylinders with a suitable hand-truck. Do not drag, slide or roll cylinders. Do not drop cylinders or permit them to strike each other. Secure cylinders firmly. Leave the valve protection cap in-place (if provided), until cylinder is ready for use.

During Use: Use designated CGA fittings and other support equipment. Do not use adapters. Do not heat cylinder by any means to increase the discharge rate of the product from the cylinder. Use check valve or trap in discharge line to prevent hazardous backflow into the cylinder. Do not use oils or grease on gas-handling fittings or equipment.

After Use: Close main cylinder valve. Replace valve protection cap (if provided). Mark empty cylinders "EMPTY".

NOTE: Use only DOT or ASME code containers. Close valve after each use and when empty. Cylinders must not be recharged except by or with the consent of owner. For additional information refer to the Compressed Gas Association Pamphlet P-1, "Safe Handling of Compressed Gases in Containers". Additionally, refer to CGA Bulletin SB-2 "Oxygen Deficient Atmospheres".

PROTECTIVE PRACTICES DURING MAINTENANCE OF CONTAMINATED EQUIPMENT: Follow practices indicated in Section 6 (Accidental Release Measures). Make certain application equipment is locked and tagged-out safely. Purge gas handling equipment with inert gas (e.g., nitrogen) before attempting repairs.

8. EXPOSURE CONTROLS - PERSONAL PROTECTION

VENTILATION AND ENGINEERING CONTROLS: Use with adequate ventilation. Local exhaust ventilation is preferred, because it prevents Tetrafluoroethane dispersion into the work place by eliminating it at its source. If necessary, the work area should be monitored for the level of oxygen.

RESPIRATORY PROTECTION: Maintain oxygen levels above 19.5% in the workplace. Use supplied air respiratory protection if oxygen levels are below 19.5% or during emergency response to a release of Tetrafluoroethane. If respiratory protection is required, follow the requirements of the Federal OSHA Respiratory Protection Standard (29 CFR 1910.134), or equivalent State standards.

EYE PROTECTION: Splash goggles or safety glasses, for protection from rapidly expanding. Face-shields should be worn if contact with the liquefied gas is anticipated.

HAND PROTECTION: Wear mechanically-resistant gloves when handling cylinders of Tetrafluoroethane. Wear Viton or rubber gloves when using Tetrafluoroethane.

BODY PROTECTION: Use body protection appropriate for task. Transfer of large quantities under pressure may require protective equipment appropriate to protect employees from splashes of liquefied Tetrafluoroethane.

9. PHYSICAL and CHEMICAL PROPERTIES

VAPOR DENSITY (air = 1): 3.6.

pH: Not applicable.

SPECIFIC GRAVITY (air = 1): 3.6

FREEZING POINT: Not determined.

SOLUBILITY IN WATER: Slightly.

BOILING POINT @ 1 atm: -26.5°C (-15.7°F)

EVAPORATION RATE (nBuAc = 1): Not applicable.

EXPANSION RATIO: Not applicable.

VAPOR PRESSURE (psia): 96

SPECIFIC VOLUME (ft³/lb): Not determined.

ODOR THRESHOLD: Not applicable.

COEFFICIENT WATER/OIL DISTRIBUTION: Not applicable.

APPEARANCE AND COLOR: Colorless, odorless, non-flammable gas.

HOW TO DETECT THIS SUBSTANCE (warning properties): There are no distinct warning properties for Tetrafluoroethane. In terms of leak detection, fittings and joints can be painted with a soap solution to detect leaks, which will be indicated by a bubble formation.

10. STABILITY and REACTIVITY

STABILITY: Stable.

DECOMPOSITION PRODUCTS: If Tetrafluoroethane is exposed to fire, it may decompose, yielding toxic products (e.g., hydrogen fluoride, carbonyl fluoride).

MATERIALS WITH WHICH SUBSTANCE IS INCOMPATIBLE: The following materials are not compatible with Tetrafluoroethane: alkaline and alkaline earth metals (such as sodium, potassium, lithium, barium, and magnesium, powdered aluminum). Metals such as silver, brass, bronze and copper may enhance the decomposition of Tetrafluoroethane at elevated temperatures. Tetrafluoroethane may also decompose in the presence of moisture and alloys which contain more than 2% magnesium.

HAZARDOUS POLYMERIZATION: Will not occur.

CONDITIONS TO AVOID: Avoid contact with incompatible materials. Cylinders exposed to high temperatures or direct flame can rupture or burst.

PART IV *Is there any other useful information about this material?*

11. TOXICOLOGICAL INFORMATION

TOXICITY DATA: The following information is available for Tetrafluoroethane.

LCLo (inhalation, rat) = 567,000 ppm/4 hours

EYE: A short duration of spray of vapor produced very slight eye irritation.

SKIN: Animal testing indicates this material is slight skin irritant, but not a skin sensitizer.

INHALATION: A single exposure caused in test rats caused cardiac sensitization, a potentially fatal disturbance of heart rhythm associated with a heightened sensitivity to the action of epinephrine. The Lowest Observable Effect Level for cardiac sensitization is 75000 ppm. Single exposure has caused the following symptoms: lethargy, narcosis, and increased respiratory rates, tremors, loss of coordination; these effects were temporary. Single exposures have also caused: increased adrenals, liver, spleen weight and decreased uterine and prostate weight.

SUSPECTED CANCER AGENT: Tetrafluoroethane is not found on the following lists: FEDERAL OSHA Z LIST, IARC, NTP, CAL/OSHA, and therefore is not considered to be, nor suspected to be a cancer-causing agent by these agencies.

IRRITANCY OF PRODUCT: Tetrafluoroethane is not irritating; however, contact with rapidly expanding gases can cause frostbite to exposed tissue.

SENSITIZATION TO THE PRODUCT: Tetrafluoroethane is not known to cause sensitization in humans upon repeated or prolonged exposure.

REPRODUCTIVE TOXICITY INFORMATION: Listed below is information concerning the effects of Tetrafluoroethane on the human reproductive system.

Mutagenicity: No mutagenicity effects on humans have been described for Tetrafluoroethane. Tests have shown that this compound does not cause genetic damage in the reproductive cells of mammals.

Embryotoxicity: No embryotoxic effects have been described for Tetrafluoroethane.

Teratogenicity: No teratogenicity effects on humans have been described for Tetrafluoroethane. Studies showed fetotoxic effects only at levels harmful to the adult animal.

Reproductive Toxicity: No reproductive toxicity effects on humans have been described for Tetrafluoroethane. Studies of mice indicate no evidence of reproductive effects after exposure.

A mutagen is a chemical which causes permanent changes to genetic material (DNA) such that the changes will propagate through generation lines. An embryotoxin is a chemical which causes damage to a developing embryo (i.e. within the first eight weeks of pregnancy in humans), but the damage does not propagate across generational lines. A teratogen is a chemical which causes damage to a developing fetus, but the damage does not propagate across generational lines. A reproductive toxin is any substance which interferes in any way with the reproductive process.

MEDICAL CONDITIONS AGGRAVATED BY EXPOSURE: Pre-existing respiratory conditions and cardiovascular conditions may be aggravated by overexposure to Tetrafluoroethane.

RECOMMENDATIONS TO PHYSICIANS: Administer oxygen, if necessary. Treat symptoms and eliminate exposure.

BIOLOGICAL EXPOSURE INDICES (BEIs): Currently, Biological Exposure Indices (BEIs) are not applicable for Tetrafluoroethane.

12. ECOLOGICAL INFORMATION

ENVIRONMENTAL STABILITY: This gas will be dissipated rapidly in well-ventilated areas.

EFFECT OF MATERIAL ON PLANTS or ANIMALS: Any adverse effect on animals would be related to adverse effects on the cardiovascular system and to exposure to oxygen-deficient environments. The symptoms experienced by over-exposed animals would be similar to those described for exposed humans. No adverse effect is anticipated to occur to plant-life, except for frost produced in the presence of rapidly expanding gases.

EFFECT OF CHEMICAL ON AQUATIC LIFE: The following Aquatic Toxicity data are available for Tetrafluoroethane:

EC₅₀ (48 hours) - *Daphnia magna*: 980 mg/L

LC₅₀ (96 hour) - Rainbow Trout: 450 mg/L

13. DISPOSAL CONSIDERATIONS

PREPARING WASTES FOR DISPOSAL: Waste disposal must be in accordance with appropriate Federal, State, and local regulations. Return cylinders with any residual product to Airgas Inc. Do not dispose of locally.

14. TRANSPORTATION INFORMATION

THIS MATERIAL IS HAZARDOUS AS DEFINED BY 49 CFR 172.101 BY THE U.S. DEPARTMENT OF TRANSPORTATION.

PROPER SHIPPING NAME: 1,1,1,2-Tetrafluoroethane

HAZARD CLASS NUMBER and DESCRIPTION: 2.2 (Nonflammable Gas)

UN IDENTIFICATION NUMBER: UN 3159

PACKING GROUP: Not Applicable

DOT LABEL(S) REQUIRED: Nonflammable Gas

NORTH AMERICAN EMERGENCY RESPONSE GUIDEBOOK NUMBER (1996): 126

MARINE POLLUTANT: Tetrafluoroethane is not classified by the DOT as a Marine Pollutant (as defined by 49 CFR 172.101, Appendix B).

TRANSPORT CANADA TRANSPORTATION OF DANGEROUS GOODS REGULATIONS: THIS MATERIAL IS CONSIDERED AS DANGEROUS GOODS. Use the above information for the preparation of Canadian Shipments.

15. REGULATORY INFORMATION

U.S. SARA REPORTING REQUIREMENTS: Tetrafluoroethane not is subject to the reporting requirements of Sections 302, 304 and 313 of Title III of the Superfund Amendments and Reauthorization Act.

U.S. SARA Threshold Planning Quantity: Not applicable.

CERCLA REPORTABLE QUANTITY (RQ): Not applicable.

CANADIAN DSL/NDL INVENTORY STATUS: Tetrafluoroethane is listed on the DSL Inventory.

U.S. TSCA INVENTORY STATUS: Tetrafluoroethane is listed on the TSCA Inventory.

OTHER U.S. FEDERAL REGULATIONS: Not applicable.

U.S. STATE REGULATORY INFORMATION: Tetrafluoroethane is not covered under specific State regulations, as denoted below:

Alaska - Designated Toxic and Hazardous Substances: No.

California - Permissible Exposure Limits for Chemical Contaminants: No.

Florida - Substance List: No.

Illinois - Toxic Substance List: No.

Kansas - Section 302/313 List: No.

Massachusetts - Substance List: No.

Michigan Critical Register List: No.

Minnesota - List of Hazardous Substances: No.

Missouri - Employer Information/Toxic Substance List: No.

New Jersey - Right to Know Hazardous Substance List: No.

North Dakota - List of Hazardous Chemicals, Reportable Quantities: No.

Pennsylvania - Hazardous Substance List: No.

Rhode Island - Hazardous Substance List: No.

Texas - Hazardous Substance List: No.

West Virginia - Hazardous Substance List: No.

Wisconsin - Toxic and Hazardous Substances: No.

CALIFORNIA SAFE DRINKING WATER AND TOXIC ENFORCEMENT ACT (PROPOSITION 65): Tetrafluoroethane is not on the California Proposition 65 lists.

15. REGULATORY INFORMATION (Continued)

LABELING:

CAUTION:

LIQUID AND GAS UNDER PRESSURE.
CAN CAUSE RAPID SUFFOCATION.
MAY CAUSE FROSTBITE.

Store and use with adequate ventilation.
Do not get liquid in eyes, on skin or clothing.
Cylinder temperature should not exceed 52°C (125°F).
Close valve after each use and when empty.
Use in accordance with the Material Safety Data Sheet.

FIRST-AID:

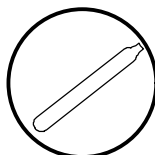
IF INHALED, remove to fresh air. If not breathing, give artificial respiration. If breathing is difficult, give oxygen. Call a physician.

IN CASE OF FROSTBITE, obtain immediate medical attention.

DO NOT REMOVE THIS PRODUCT LABEL.

CANADIAN WHMIS SYMBOLS:

Class A: Compressed Gases



16. OTHER INFORMATION

PREPARED BY:

CHEMICAL SAFETY ASSOCIATES, Inc.
9163 Chesapeake Drive, San Diego, CA 92123-1002
619/565-0302

The information contained herein is based on data considered accurate. However, no warranty is expressed or implied regarding the accuracy of these data or the results to be obtained from the use thereof. AIRGAS, Inc. assumes no responsibility for injury to the vendee or third persons proximately caused by the material if reasonable safety procedures are not adhered to as stipulated in the data sheet. Additionally, AIRGAS, Inc. assumes no responsibility for injury to vendee or third persons proximately caused by abnormal use of the material even if reasonable safety procedures are followed. Furthermore, vendee assumes the risk in his use of the material.

DEFINITIONS OF TERMS

A large number of abbreviations and acronyms appear on a MSDS. Some of these which are commonly used include the following:

CAS #: This is the Chemical Abstract Service Number which uniquely identifies each constituent. It is used for computer-related searching.

EXPOSURE LIMITS IN AIR:

ACGIH - American Conference of Governmental Industrial Hygienists, a professional association which establishes exposure limits. **TLV** - Threshold Limit Value - an airborne concentration of a substance which represents conditions under which it is generally believed that nearly all workers may be repeatedly exposed without adverse effect. The duration must be considered, including the 8-hour Time Weighted Average (**TWA**), the 15-minute Short Term Exposure Limit, and the instantaneous Ceiling Level (**C**). Skin absorption effects must also be considered.

OSHA - U.S. Occupational Safety and Health Administration. **PEL** - Permissible Exposure Limit - This exposure value means exactly the same as a TLV, except that it is enforceable by OSHA. The OSHA Permissible Exposure Limits are based in the 1989 PELs and the June, 1993 Air Contaminants Rule (*Federal Register*: 58: 35338-35351 and 58: 40191). Both the current PELs and the vacated PELs are indicated.

The phrase, "Vacated 1989 PEL," is placed next to the PEL which was vacated by Court Order.

IDLH - Immediately Dangerous to Life and Health - This level represents a concentration from which one can escape within 30-minutes without suffering escape-preventing or permanent injury. **The DFG - MAK** is the Republic of Germany's Maximum Exposure Level, similar to the U.S. PEL. **NIOSH** is the National Institute of Occupational Safety and Health, which is the research arm of the U.S. Occupational Safety and Health Administration (**OSHA**). NIOSH issues exposure guidelines called Recommended Exposure Levels (**RELs**). When no exposure guidelines are established, an entry of **NE** is made for reference.

HAZARD RATINGS:

HAZARDOUS MATERIALS IDENTIFICATION SYSTEM: Health Hazard: **0** (minimal acute or chronic exposure hazard); **1** (slight acute or chronic exposure hazard); **2** (moderate acute or significant chronic exposure hazard); **3** (severe acute exposure hazard; onetime overexposure can result in permanent injury and may be fatal); **4** (extreme acute exposure hazard; onetime overexposure can be fatal). Flammability Hazard: **0** (minimal hazard); **1** (materials that require substantial pre-heating before burning); **2** (combustible liquid or solids; liquids with a flash point of 38-93°C [100-200°F]); **3** (Class IB and IC flammable liquids with flash points below 38°C [100°F]); **4** (Class IA flammable liquids with flash points below 23°C [73°F] and boiling points below 38°C [100°F]). Reactivity Hazard: **0** (normally stable); **1** (material that can become unstable at elevated temperatures or which can react slightly with water); **2** (materials that are unstable but do not detonate or which can react violently with water); **3** (materials that can detonate when initiated or which can react explosively with water); **4** (materials that can detonate at normal temperatures or pressures).

NATIONAL FIRE PROTECTION ASSOCIATION: Health Hazard: **0** (material that on exposure under fire conditions would offer no hazard beyond that of ordinary combustible materials); **1** (materials that on exposure under fire conditions could cause irritation or minor residual injury); **2** (materials that on intense or continued exposure under fire conditions could cause temporary incapacitation or possible residual injury); **3** (materials that can on short exposure could cause serious temporary or residual injury); **4** (materials that under very short exposure causes death or major residual injury). Flammability Hazard and Reactivity Hazard: Refer to definitions for "Hazardous Materials Identification System".

FLAMMABILITY LIMITS IN AIR:

Much of the information related to fire and explosion is derived from the National Fire Protection Association (**NFPA**). Flash Point - Minimum temperature at which a liquid gives off sufficient vapors to form an ignitable mixture with air. Autoignition Temperature: The minimum temperature required to initiate combustion in air with no other source of ignition. LEL - the lowest percent of vapor in air, by volume, that will explode or ignite in the presence of an ignition source. UEL - the highest percent of vapor in air, by volume, that will explode or ignite in the presence of an ignition source.

TOXICOLOGICAL INFORMATION:

Possible health hazards as derived from human data, animal studies, or from the results of studies with similar compounds are presented. Definitions of some terms used in this section are: **LD₅₀** - Lethal Dose (solids & liquids) which kills 50% of the exposed animals; **LC₅₀** - Lethal Concentration (gases) which kills 50% of the exposed animals; **ppm** concentration expressed in parts of material per million parts of air or water; **mg/m³** concentration expressed in weight of substance per volume of air; **mg/kg** quantity of material, by weight, administered to a test subject, based on their body weight in kg. Data from several sources are used to evaluate the cancer-causing potential of the material. The sources are: **IARC** - the International Agency for Research on Cancer; **NTP** - the National Toxicology Program, **RTECS** - the Registry of Toxic Effects of Chemical Substances, **OSHA** and **CAL/OSHA**. IARC and NTP rate chemicals on a scale of decreasing potential to cause human cancer with rankings from 1 to 4. Subrankings (2A, 2B, etc.) are also used. Other measures of toxicity include **TDLo**, the lowest dose to cause a symptom and **TCLo** the lowest concentration to cause a symptom; **TD₀₁**, **LDLo**, and **LD₀₁**, or **TC**, **TC₀₁**, **LCLo**, and **LC₀₁**, the lowest dose (or concentration) to cause lethal or toxic effects. **BEI** - Biological Exposure Indices, represent the levels of determinants which are most likely to be observed in specimens collected from a healthy worker who has been exposed to chemicals to the same extent as a worker with inhalation exposure to the TLV. Ecological Information: EC is the effect concentration in water.

REGULATORY INFORMATION:

This section explains the impact of various laws and regulations on the material. **EPA** is the U.S. Environmental Protection Agency. **WHMIS** is the Canadian Workplace Hazardous Materials Information System. **DOT** and **TC** are the U.S. Department of Transportation and the Transport Canada, respectively. Superfund Amendments and Reauthorization Act (**SARA**); the Canadian Domestic/Non-Domestic Substances List (**DSL/NDSL**); the U.S. Toxic Substance Control Act (**TSCA**); Marine Pollutant status according to the **DOT**; the Comprehensive Environmental Response, Compensation, and Liability Act (**CERCLA** or **Superfund**); and various state regulations.

Material Safety Data Sheet

Midland, Michigan 48674

Emergency 517 - 636-4400

1. CHEMICAL PRODUCT & COMPANY IDENTIFICATION

Page: 1

24-Hour Emergency Phone Number: 989-636-4400

Product: DOWFROST* HEAT TRANSFER FLUID

Product Code: 23545

Effective Date: 01/08/99 Date Printed: 05/07/01 MSD: 000130

The Dow Chemical Company, Midland, MI 48674

Customer Information Center: 800-258-2436

2. COMPOSITION/INFORMATION ON INGREDIENTS

Propylene glycol CAS# 000057-55-6 95%

Dipotassium phosphate CAS# 007758-11-4 <5%

Deionized water CAS# 007732-18-5 <5%

3. HAZARDS IDENTIFICATION

EMERGENCY OVERVIEW

* Colorless, odorless liquid. Toxic fumes released in fire * situations.

POTENTIAL HEALTH EFFECTS (See Section 11 for toxicological data.)

EYE: May cause slight transient eye irritation. Corneal injury is unlikely.

SKIN: Essentially nonirritating to skin on prolonged contact. A single prolonged skin exposure is not likely to result in the material being absorbed through skin in harmful amounts. Repeated exposures may cause slight flaking, tenderness and softening of skin.

(Continued on page 2)

OR (R) INDICATES A TRADEMARK OF THE DOW CHEMICAL COMPANY

Printed on Recycled and Recyclable Paper

low

Product: DOWFROST* HEAT TRANSFER FLUID Product Code: **23545**

Effective Date: 01/08/99 Date Printed: 05/07/01 MSD: **000130**

INGESTION: Single dose oral toxicity is low. No hazards anticipated from swallowing small amounts incidental to normal handling operations.

INHALATION: A single prolonged (hours) inhalation exposure is not likely to cause adverse effects. Mists are not likely to be hazardous.

SYSTEMIC (OTHER TARGET ORGAN) EFFECTS: Repeated excessive ingestion may cause central nervous system effects.

CANCER INFORMATION: Did not cause cancer in long-term animal studies.

TERATOLOGY (BIRTH DEFECTS): Birth defects are unlikely. Exposures having no adverse effects on the mother should have no effect on the fetus.

REPRODUCTIVE EFFECTS: In animal studies, has been shown not to interfere with reproduction.

4. FIRST AID

EYE: Flush eyes with plenty of water.

SKIN: Wash off in flowing water or shower.

INGESTION: If swallowed, seek medical attention. Do not induce vomiting unless directed to do so by medical personnel.

INHALATION: Remove to fresh air if effects occur. Consult a physician.

NOTE TO PHYSICIAN: No specific antidote. Supportive care. Treatment based on judgment of the physician in response to reactions of the patient.

5. FIRE FIGHTING MEASURES

FLASH POINT: 215F, 102C **METHOD USED:** TOC (Tag Open Cup)

(Continued on page 3)

* OR (R) INDICATES A TRADEMARK OF THE DOW CHEMICAL COMPANY

Product: DOWFROST* HEAT TRANSFER FLUID

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FLAMMABLE LIMITS LFL: 2.6% @ 100C UFL: 12.5% @ 130C

HAZARDOUS COMBUSTION PRODUCTS: During a fire, smoke may contain the original material in addition to unidentified toxic and/or irritating compounds. Hazardous combustion products may include and are not limited to carbon monoxide, carbon dioxide.

OTHER FLAMMABILITY INFORMATION: Violent steam generation or eruption may occur upon application of direct water stream. Flammable concentration of vapor can accumulate at temperatures above 215.0 deg. F. Liquid mist of this product can burn. Spills of these organic liquids on hot fibrous insulations may lead to lowering of the autoignition temperatures possibly resulting in spontaneous combustion.

EXTINGUISHING MEDIA: Water fog or fine spray, carbon dioxide, dry chemical, foam. Alcohol resistant foams (ATC type) are preferred if available. General purpose synthetic foams (including AFFF) or protein foams may function, but much less effectively. Do not use direct water stream. Will spread fire.

MEDIA TO BE AVOIDED: Do not use direct water stream.

FIRE FIGHTING INSTRUCTIONS: Keep people away. Isolate fire area and deny unnecessary entry. Burning liquids may be moved by flushing with water to protect personnel and minimize property damage. Burning liquids may be extinguished by dilution with water. Do not use direct water stream. May spread fire.

PROTECTIVE EQUIPMENT FOR FIRE FIGHTERS: Wear positive-pressure self-contained breathing apparatus (SCBA) and protective firefighting clothing (includes fire-fighting helmet, coat, pants, boots, and gloves). If protective equipment is not available or not used, fight fire from a protected location or safe distance.

6. ACCIDENTAL RELEASE MEASURES (See Section 15 for Regulatory Information)

(Continued on page 4)

* OR (R) INDICATES A TRADEMARK OF THE DOW CHEMICAL COMPANY

Product: DOWFROST* HEAT TRANSFER FLUID

Product Code: 23545

Effective Date: 01/08/99 Date Printed: 05/07/01 MSD: 000130

PROTECT PEOPLE: Clear non-emergency personnel from area.

PROTECT THE ENVIRONMENT: Contain liquid to prevent contamination of soil, surface water or ground water.

CLEANUP: Clean up with absorbent material. Sweep up.

7. HANDLING AND STORAGE

HANDLING: Product on surfaces can cause slippery conditions.

STORAGE: Store below 121 C, 250 F.

8. EXPOSURE CONTROLS/PERSONAL PROTECTION

ENGINEERING CONTROLS: Good general ventilation should be sufficient for most conditions.

PERSONAL PROTECTIVE EQUIPMENT

EYE/FACE PROTECTION: Use safety glasses.

SKIN PROTECTION: For brief contact, no precautions other than clean body-covering clothing should be needed. Use impervious gloves when prolonged or frequently repeated contact could occur.

RESPIRATORY PROTECTION: No respiratory protection should be needed.

EXPOSURE GUIDELINE(S): Propylene glycol: AIHA WEEL is 50 ppm total, 10 mg/m³ aerosol only.

9. PHYSICAL AND CHEMICAL PROPERTIES

APPEARANCE: Colorless. ODOR: Odorless liquid. VAPOR PRESSURE: 0.22 mmHg @ 20C, 68F
VAPOR DENSITY: 2.62 BOILING POINT: 370F, 188C SOLUBILITY IN WATER: Complete

(Continued on page 5)

* OR (R) INDICATES A TRADEMARK OF THE DOW CHEMICAL COMPANY

Product: DOWFROST* HEAT TRANSFER FLUID

Product Code: 23545

Effective Date: 01/08/99 Date Printed: 05/07/01 MSD: 000130

SPECIFIC GRAVITY: 1.050 @ 60/60F, 16C

10. STABILITY AND REACTIVITY

CHEMICAL STABILITY: Thermally stable at typical use temperatures.

CONDITIONS TO AVOID: Avoid temperatures above 121C/250F. Product can decompose at elevated temperatures.

INCOMPATIBILITY WITH OTHER MATERIALS: Avoid contact with oxidizing materials.

HAZARDOUS DECOMPOSITION PRODUCTS: Hazardous decomposition products depend upon temperature, air supply and the presence of other materials.

HAZARDOUS POLYMERIZATION: Will not occur.

11. TOXICOLOGICAL INFORMATION (See Section 3 for Potential Health Effects. For detailed toxicological data, write or call the address or non-emergency number shown in Section 1)

SKIN: The LD50 for skin absorption in rabbits is greater than 10,000 mg/kg.

INGESTION: The oral LD50 for female rats is about 20.3 g/kg.

MUTAGENICITY (EFFECTS ON GENETIC MATERIAL): Results of in vitro (test tube) mutagenicity tests have been negative. Results of mutagenicity tests in animals have been negative.

12. ECOLOGICAL INFORMATION (For detailed Ecological data, write or call the address or non-emergency number shown in Section 1)

ENVIRONMENTAL FATE MOVEMENT AND PARTITIONING: Based largely or completely on data for major component(s). Bioconcentration potential is low (BCF less than 100 or Log Pow less than 3).

DEGRADATION AND PERSISTENCE: Based largely or completely on data for major component(s). Biodegradation under aerobic static laboratory conditions is high (BOD20 or BOD28/ThOD

(Continued on page 6)

* OR (R) INDICATES A TRADEMARK OF THE DOW CHEMICAL COMPANY

Product: DOWFROST* HEAT TRANSFER FLUID

Product Code: 23545

Effective Date: 01/08/99 Date Printed: 05/07/01 MSD: 000130

greater than 40%).

ECOTOXICITY: Based largely or completely on data for major component(s). Material is practically non-toxic to aquatic organisms on an acute basis (LC50 greater than 100 mg/L in most sensitive species).

13. DISPOSAL CONSIDERATIONS (See Section 15 for Regulatory Information)

DISPOSAL: DO NOT DUMP INTO ANY SEWERS, ON THE GROUND OR INTO ANY BODY OF WATER. All disposal methods must be in compliance with all Federal, State/Provincial and local laws and regulations. Regulations may vary in different locations. Waste characterizations and compliance with applicable laws are the responsibility solely of the waste generator. THE DOW CHEMICAL COMPANY HAS NO CONTROL OVER THE MANAGEMENT PRACTICES OR MANUFACTURING PROCESSES OF PARTIES HANDLING OR USING THIS MATERIAL. THE INFORMATION PRESENTED HERE PERTAINS ONLY TO THE PRODUCT AS SHIPPED IN ITS INTENDED CONDITION AS DESCRIBED IN MSDS SECTION 2 (Composition/information On Ingredients).

FOR UNUSED & UNCONTAMINATED PRODUCT, the preferred options include sending to a licensed, permitted: recycler, reclaimer, or waste water treatment system.

As a service to its customers, Dow can provide names of information resources to help identify waste management companies and other facilities which recycle, reprocess or manage chemicals or plastics, and that manage used drums. Telephone Dow's Customer Information Center at 800-258-2436 or 517-832-1556 for further details.

14. TRANSPORT INFORMATION

CANADIAN TDG INFORMATION: For TDG regulatory information, if required, consult transportation regulations, product shipping papers, or your Dow representative.

15. REGULATORY INFORMATION (Not meant to be all-inclusive- -selected regulations represented)

NOTICE: The information herein is presented in good faith and

(Continued on page 7)

* OR (R) INDICATES A TRADEMARK OF THE DOW CHEMICAL COMPANY

Product: DOWFROST* HEAT TRANSFER FLUID

Product Code: 23545

Effective Date: **01/08/99** Date Printed: 05/07/01 MSD: 000130

believed to be accurate as of the effective date shown above. However, no warranty, express or implied is given. Regulatory requirements are subject to change and may differ from one location to another; it is the buyer's responsibility to ensure that its activities comply with federal, state or provincial, and local laws. The following specific information is made for the purpose of complying with numerous federal, state or provincial, and local laws and regulations. See other sections for health and safety information.

U.S. REGULATIONS

SARA 313 INFORMATION: To the best of our knowledge, this product contains no chemical subject to SARA Title III Section 313 supplier notification requirements.

SARA HAZARD CATEGORY: This product has been reviewed according to the

EPA "Hazard Categories" promulgated under Sections 311 and 312 of the Superfund Amendment and Reauthorization Act of **1986** (SARA Title 111) and is considered, under applicable definitions, to meet the following categories:

Not to have met any hazard category

TOXIC SUBSTANCES CONTROL ACT (TSCA):

All ingredients are on the TSCA inventory or are not required to be listed on the TSCA inventory.

STATE RIGHT-TO-KNOW: The following product components are cited on certain state lists as mentioned. Non-listed components may be shown in the composition section of the MSDS.

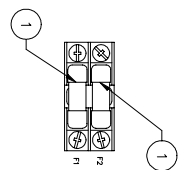
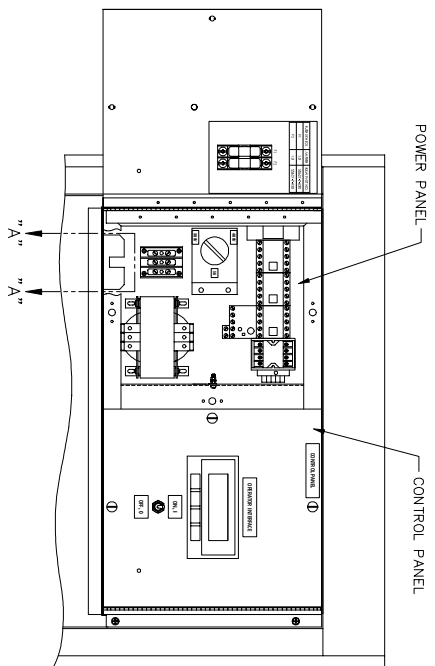
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* OR (R) INDICATES A TRADEMARK OF THE DOW CHEMICAL COMPANY

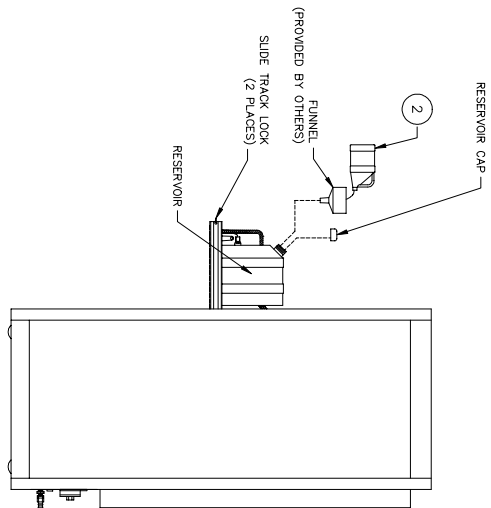
REV	DESCRIPTION	REVISIONS		BY	CHECKED	APPROVED	ENGINEERING	Q.A.
		DATE (MM-DD-YY)	DATE (MM-DD-YY)					
A	REVISED PER EDN #1-79	01-06-79		WFR	01-06-79	01-06-79	01-06-79	01-06-79
B	REVISED PER EDN #1-101	01-08-08		WFR	01-08-08	01-08-08	01-08-08	01-08-08

NOTES:

1. FUSE REPLACEMENT:
A. DISCONNECT AND LOCKOUT/TAGOUT PRIMARY POWER AT THE EXTERNAL DISCONNECT.
B. OPEN ELECTRICAL ACCESS PANEL.
C. REMOVE FUSE WITH A PAIR OF FUSE PULLERS.
D. PERFORM A CONTINUITY CHECK WITH AN OHM METER. DISCARD DEFECTIVE FUSE.
E. INSTALL NEW FUSE WITH A PAIR OF FUSE PULLERS. ENSURE REPLACEMENT FUSE IS THE SAME TYPE (AMP, VOLTAGE, ETC.) AS THE ORIGINAL.
F. CLOSE ELECTRICAL ACCESS PANEL.
G. REMOVE LOCKOUT/TAGOUT AND APPLY POWER.
H. PERFORM START UP PROCEDURE AS PER SECTION 3 OF THE OPERATION AND MAINTENANCE MANUAL.
2. RESERVOIR COOLANT FILLING:
A. UNLOCK AND OPEN FRONT ACCESS DOOR.
B. UNLOCK SLIDE TRACKS AND PULL TRAY OUT TO EXPOSE RESERVOIR.
C. REMOVE THE RESERVOIR CAP.
D. PLACE FUNNEL IN THE RESERVOIR. (FUNNEL SUPPLIED BY OTHERS)
E. ADD COOLANT AS REQUIRED TO OBTAIN A FULL READING.
F. REMOVE FUNNEL.
G. REINSTALL RESERVOIR CAP.
H. SLIDE TRAY INTO CABINET UNTIL SLIDE TRACKS LOCK.
I. CLOSE AND LOCK FRONT ACCESS DOOR.



ELECTRICAL BOX ASSEMBLY

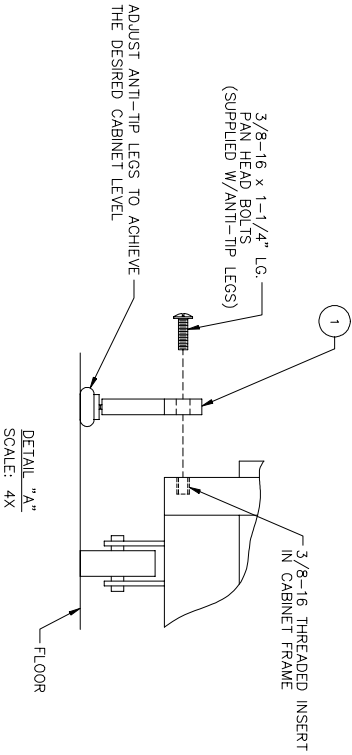
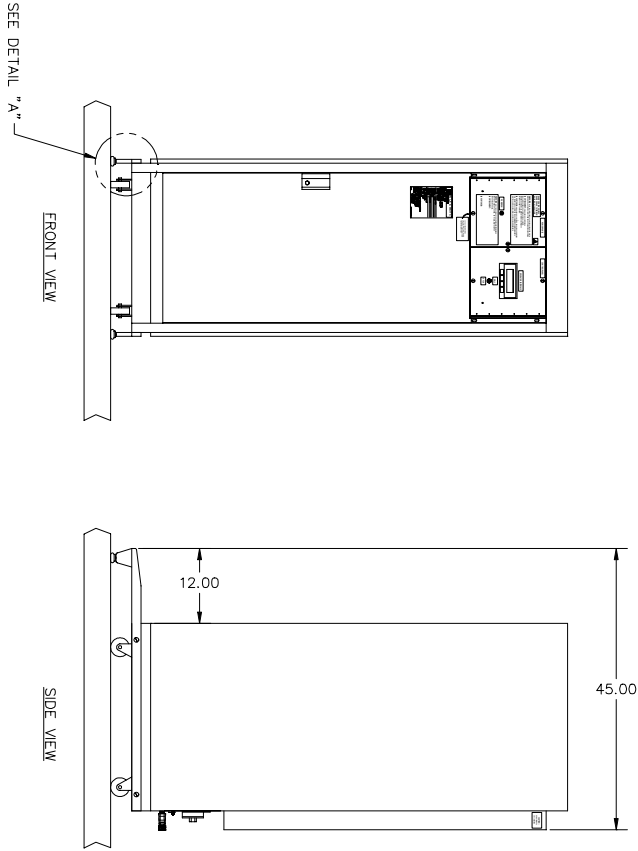


RESERVOIR COOLANT FILLING ILLUSTRATION

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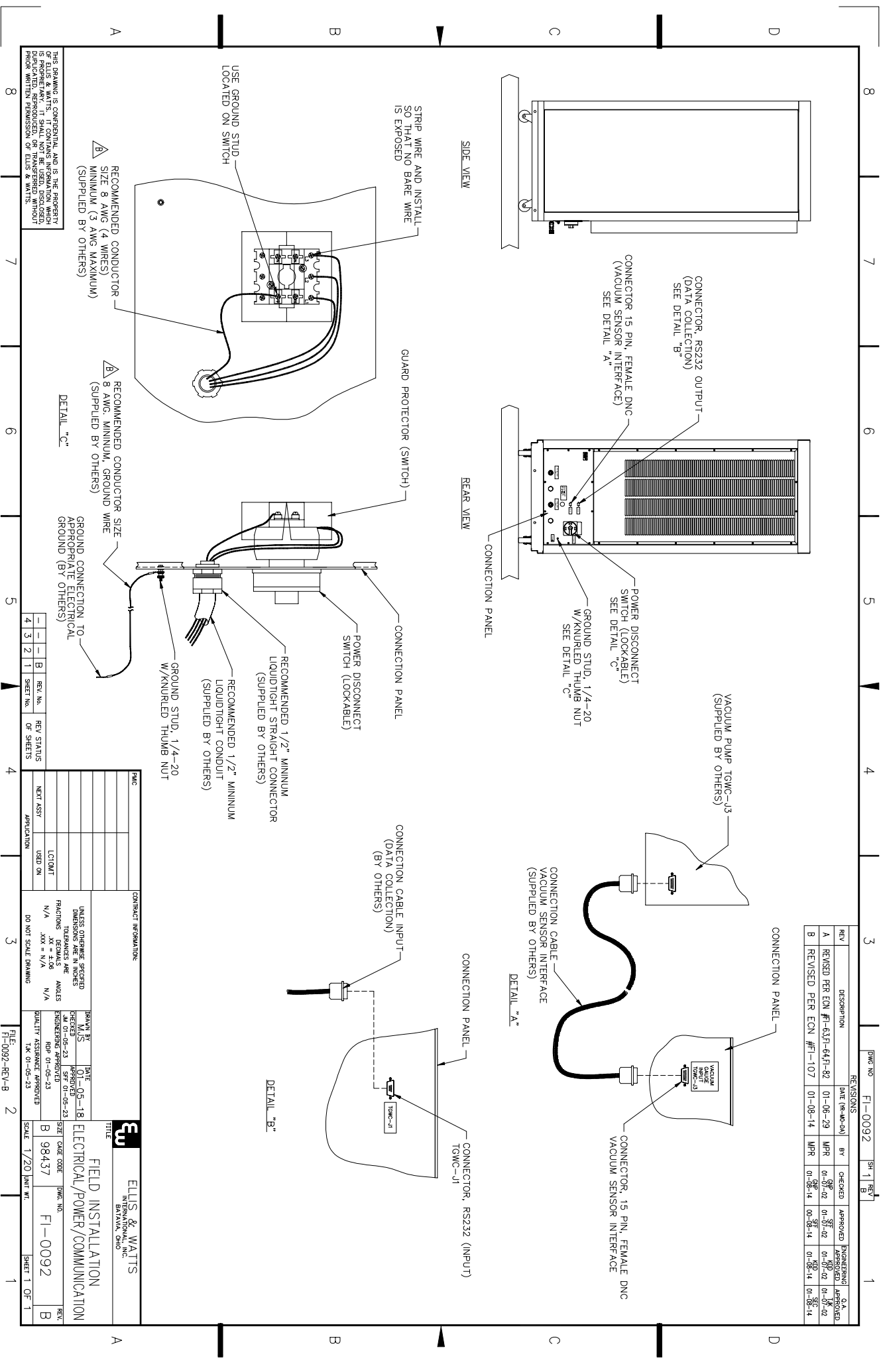
CONTRACT INFORMATION:		TITLE	
P/C		ELLIS & WATTS	FIELD INSTALLATION
		INTERNATIONAL, INC.	SPARE PARTS KIT
		BRANNA, OHIO	
UNLESS OTHERWISE SPECIFIED		DRAWN BY	
DIMENSIONS ARE IN INCHES		A/G	
FRACTIONS DECIMALS ANGLES		CHECKED BY	
N/A .XX ± .06 N/A		01-06-06	
DO NOT SCALE DRAWING		ENGINEERING APPROVED	
		KDD 01-06-07	
		QUALITY ASSURANCE APPROVED	
		SEC 01-06-07	
		SCALE N.T.S.	
		DWG. NO.	
		B 98437	
		FI-0097	
		SHEET 1 OF 1	

REV		DESCRIPTION	DATE (YY-MM-DD)	BY	CHECKED	APPROVED	ENGINEERING	APPROVED	O.A.
A	REVISED PER ECN #1-09		01-03-12	MJS	01-03-20	01-03-23	01-03-23	01-03-23	01-03-23
B	REVISED PER ECN #1-31		01-05-17	MJS	01-05-18	01-05-18	01-05-18	01-05-18	01-05-18
C	REVISED PER ECN #1-47		01-05-23	MJS	01-05-23	01-05-23	01-05-23	01-05-23	01-05-23
D	REVISED PER ECN #1-61		01-06-04	AJC	01-06-04	01-06-05	01-06-05	01-06-05	01-06-05



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NO.		PART NO.	ANTI-TIP LEG W/MOUNTING HARDWARE	QTY.
1		T2624-172A		2
CONTRACT INFORMATION		SHIP LOOSE ITEMS DESCRIPTION		
UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN INCHES. FRACTIONS: DECIMALS: ANGLES: TOLERANCES ARE: .XX = ±.06 N/A N/A N/A		DRAWN BY: MJS DATE: 01-02-13		
DO NOT SCALE DRAWING		CHECKED BY: MJS DATE: 01-02-21		
NEXT ASSY		ENGINEERING APPROVED: KDD 01-02-21		
APPLICATION		QUALITY ASSURANCE APPROVED: RRS 01-02-22		
TITLE		FIELD INSTALLATION		
ELLIS & WATTS		ANTI-TIP LEGS		
BATAVA, OHIO		DWG. NO. FI-0014		
SCALE 1/20		UNIT WT.		
SHEET 1 OF 1		REV D		



8	7	6	5	4	3	2	1
4	3	2	1				
REV. NO.	REV. STATUS	REV. NO.	REV. STATUS	REV. NO.	REV. STATUS	REV. NO.	REV. STATUS
B	USED ON	B	USED ON	B	USED ON	B	USED ON
3	USED ON	3	USED ON	3	USED ON	3	USED ON
2	USED ON	2	USED ON	2	USED ON	2	USED ON
1	USED ON	1	USED ON	1	USED ON	1	USED ON

FILE: 0092-REV-B 2 1

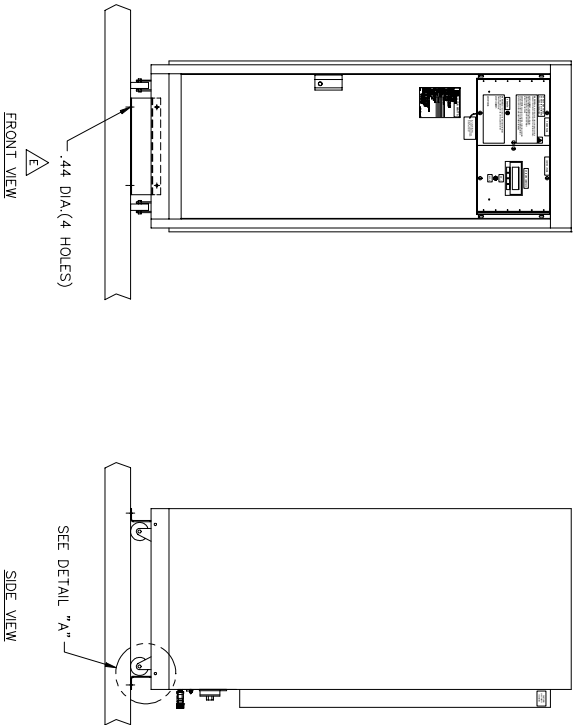
DWG NO. FI-0017			REV			1		
REVISIONS			SH			E		
REV	DESCRIPTION	DATE (MM-DD-YY)	BY	CHECKED	APPROVED	ENGINEERING	O.A.	APPROVED
A	REVISED PER ECN #1-08	01-03-12	MJS	01-06-20	01-13-23	01-06-23	01-06-23	01-06-23
B	REVISED PER ECN #1-32	01-05-17	MJS	01-06-18	01-06-18	01-06-18	01-06-18	01-06-18
C	REVISED PER ECN #1-46	01-05-23	MJS	01-06-23	01-06-23	01-06-23	01-06-23	01-06-23
D	REVISED PER ECN #1-59	01-06-04	MJS	01-06-04	01-06-04	01-06-04	01-06-04	01-06-04
E	REVISED PER ECN #1-104	01-08-10	MJR	01-08-10	01-08-10	01-08-10	01-08-10	01-08-10

NOTES

1. SLIDE TOP OF BRACKET INTO CABINET.
THEN ROTATE INTO POSITION.

2. ATTACH EACH BRACKET WITH (2) 1/4-20 x 1-1/4" LG.
TRUSS HEAD SCREWS (SUPPLIED BY E&W).

3. USE SEISMIC TIE-DOWN BRACKETS WHERE APPLICABLE.



FRONT VIEW

SIDE VIEW

SEE DETAIL "A"

FRONT OF UNIT

SEISMIC TIE-DOWN BRACKET HOLE PATTERN

NO.	PART NO.	SHIP LOOSE ITEMS DESCRIPTION	QTY.
4	50-22-126	NUT, LOCK, 1/4-20 SST	4
3	50-3E-34	WASHER, FLAT, 1/4" SST	4
2	50-19X-633	SCREW, TRUSS HEAD PHILLIPS, 1/4-20 x 1-1/4" LG. SST	4
1	T2624-0201-21	SEISMIC TIE-DOWN BRACKET	2

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DETAIL "A"

SCALE: 4X

REV	DESCRIPTION	DATE (MM-DD-YY)	BY	CHECKED	APPROVED	ENGINEERING	O.A.	APPROVED
4	3	2	1	1	1	1	1	1

CONTRACT INFORMATION			FIELD INSTALLATION		
PAC	APPLICATION	USED ON	UNLESS OTHERWISE SPECIFIED	DATE	BY
			FRACIONS ARE IN INCHES	01-02-14	MJS
			RELATIVES ARE ANGLES	01-02-21	01-02-21
			N/A	01-02-21	01-02-21
			DO NOT SCALE DRAWING	01-02-22	01-02-22
			SCALE	1/20	DATE
			FILE:FI-0017-REV-E-2	1	1

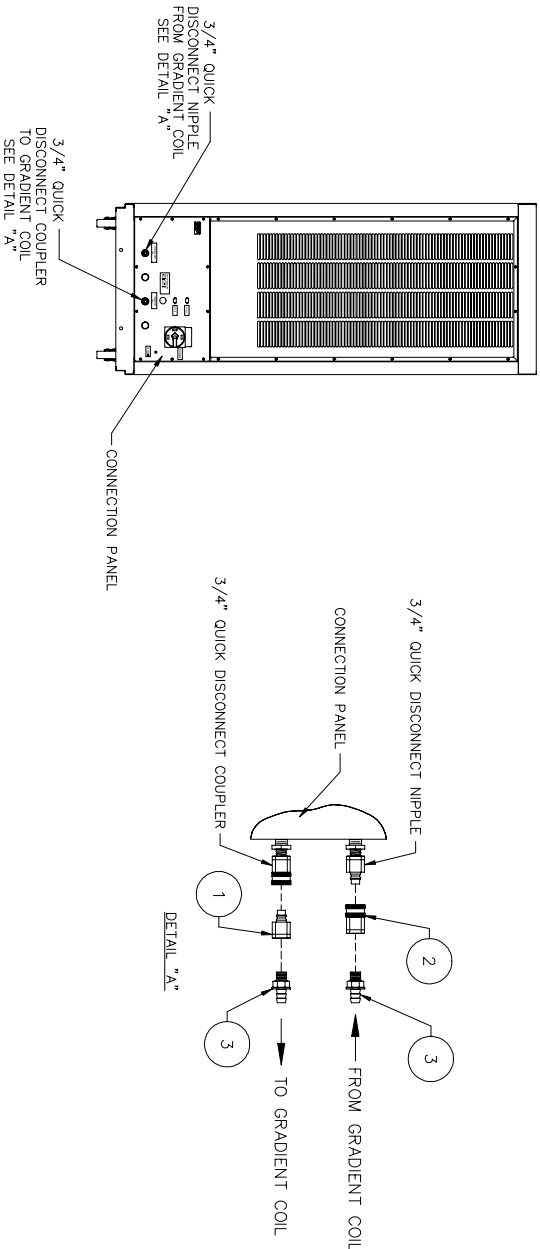
FILE:FI-0017-REV-E-2

SHEET 1 OF 1

REV		DESCRIPTION		DATE (YY-MM-DD)		BY		CHECKED		APPROVED		REVISIONS	
A	REVISED PER ECU #1-62	01-06-04	AIG	01-06-04	01-06-05	01-06-05	01-06-05	01-06-05	01-06-05	01-06-05	01-06-05	01-06-05	01-06-05
B	REVISED PER ECU #1-117	01-09-10	WPR	01-09-10	01-09-10	01-09-10	01-09-10	01-09-10	01-09-10	01-09-10	01-09-10	01-09-10	01-09-10

NOTES:
1. USE LOCITITE 567 HIGH TEMPERATURE THREAD SEALANT OR EQUAL ON ALL THREADED FITTINGS.

2. TO ASSEMBLE PUSH-LOK HOSE FITTINGS:
A. CUT HOSE CLEANLY AND SQUARELY WITH A SHARP KNIFE.
B. LUBRICATE THE PUSH-LOK FITTING AND OR HOSE I.D. WITH A LIGHT COATING OF GYLCOOL WATER SOLUTION.
C. INSERT FITTING INTO HOSE UNTIL FIRST BARB IS IN HOSE.
D. PLACE END FITTING AGAINST A FLAT OBJECT (BENCH, WALL). GRIP HOSE APPROXIMATELY ONE INCH FROM END AND PUSH WITH STEADY FORCE UNTIL THE END OF THE HOSE IS COVERED BY THE YELLOW PLASTIC CAP.
3. TO REPLACE PUSH-LOK HOSE:
A. LEAVE FITTING IN PLACE, AND CUT HOSE LENGTH-WISE FROM THE YELLOW PLASTIC CAP APPROXIMATELY ONE INCH. IMPORTANT: BE CAREFUL NOT TO NICK BARBS WHEN CUTTING HOSE.
B. GRIP HOSE AND GIVE A SHARP DOWNWARD TUG TO DISENGAGE HOSE FROM FITTING.



3	J1476-206	3/4" HOSE BARB, 3/4" NPT MALE	2
2	T2637-140ZS	3/4" QUICK DISCONNECT COUPLER, 3/4" NPT FEMALE	1
1	T2637-142ZS	3/4" QUICK DISCONNECT NIPPLE, 3/4" NPT FEMALE	1
NO.		PART NO.	SHIP LOOSE ITEMS DESCRIPTION

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4	3	2	1	B	REV. NO.	REV. STATUS
4	3	2	1	B	REV. NO.	REV. STATUS

CONTRACT INFORMATION		FIELD INSTALLATION	
PAC		TITLE	
UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN INCHES		DATE	
TOLERANCES ARE		CHECKED	
FRACTIONS XX' = N/A		DATE	
DO NOT SCALE DRAWING		DATE	
QUALITY ASSURANCE APPROVED		DATE	
SCALE 1/2"		DATE	
SHEET 1 OF 1		DATE	